

# *PSOPT* Application Examples Document

Victor Becerra  
Email: [v.m.becerra@ieee.org](mailto:v.m.becerra@ieee.org)  
www: <http://www.psopt.net>  
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# *PSOPT* Application Examples

Victor Becerra  
Email: v.m.becerra@ieee.org

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## 1 About this document

This document presents a small set of optimal control problems solved with *PSOPT*. It is not a catalogue of everything the distribution contains: the **examples** directory holds several dozen problems, and the purpose here is narrower. Each problem below was chosen to stand for one class of problem or one capability of the software, so that a reader who has a problem of their own can find the nearest relative, read how it was posed, and then go to the source file and copy the parts that apply. The classes covered are a single-phase problem with free final time, a multi-phase problem, parameter estimation, a problem with a path constraint, a problem with an interior-point constraint, a singular problem, delayed states, a high-index differential-algebraic system, an integer static parameter, an integer control, the integrated-residual transcription, a chance-constrained problem, automatic mesh refinement in both its local and its *hp*-adaptive forms, a minimum-time problem driven by tabulated data, and a multi-phase launch vehicle trajectory.

Each section has the same three parts. A short discussion says what the problem is, where it comes from and what is worth noticing about solving it. A summary table gives the problem in the form *PSOPT* asks for it: the objective, the dynamics, the boundary conditions, the path constraints, the bounds, the constants, and the discretisation and mesh refinement settings used. Then the results: the solver's own summary, the figures, and, where the problem has a published or analytic answer, a comparison against it.

The summary tables are meant to be read on their own. Where a problem's data are bulky rather than interesting (a table of aerodynamic coefficients, a list of measurements, a set of physical constants), the table says where they are rather than reproducing them, and the reader is referred to the source file. The complete, runnable source of every example is in the *PSOPT* distribution, under **examples/<name>/**. Source listings are printed here for three of the simplest problems only, in the expectation that a reader who has understood how those are put together will prefer to open the others in an editor.

Most of these problems are taken from the literature, so that the solutions computed here can be checked against published ones. Where such a value exists it is quoted in the summary table and compared with what *PSOPT* produces. Where it does not, because

the problem is new or because its answer is an infimum rather than a minimum, the section says so.

## Reproducing the results

Everything numerical in this document is generated rather than transcribed. The figures, the results boxes, the mesh refinement tables, the parameter statistics and the source listings are files under `generated/`, produced by the script `tools/regenerate.sh` from the release of *PSOPT* recorded in `tools/PSOPT_VERSION`. Running

```
tools/regenerate.sh all
```

clones and builds that release, runs every example in the document, and refreshes `generated/`. A reader who wants to check a number quoted in the text can find the run it came from in `generated/<example>/`.

PSOPT release 2026-08-20-63-g<climb-us76>, built 1 September 2026. Results regenerated on 01 September 2026.

Computing times are reported as they were measured and should be read as indicative only, since they depend on the machine; the node counts, iteration counts, error estimates and objective values do not.

## 2 Brachistochrone: a single-phase problem with free final time

The brachistochrone is the oldest problem in the calculus of variations: given two points in a vertical plane, find the shape of the wire down which a bead, released from rest and sliding without friction, travels between them in the least time. Johann Bernoulli posed it in 1696 and the answer, that the curve is a cycloid, was supplied within a year by Newton, Leibniz, l'Hôpital, Jacob Bernoulli and Johann Bernoulli himself. It is used here as the simplest possible complete example: one phase, three states, one control, a free final time, no path constraints, and an answer that can be written down in closed form and compared with what the software produces.

Posed as an optimal control problem [4], the control is the angle  $\theta$  of the wire from the vertical, the states are the two coordinates of the bead and its speed, and the cost is the final time itself. Measuring  $y$  downwards makes the dynamics free of a sign convention: the bead accelerates at  $g \cos \theta$  along the wire, and conservation of energy, which the equations imply rather than impose, gives  $v = \sqrt{2gy}$  along any trajectory that starts from rest at the origin.

Two features of the formulation are worth noticing because they recur in every later example. The final time is free, and is made so by giving it a bound ( $t_f \in [0, 10]$ ) rather than a value; *PSOPT* then treats it as one more decision variable. And the terminal conditions are imposed as event constraints, with equal lower and upper bounds, which is how an equality is written throughout the interface.

Because the problem is smooth and has no active path constraint, a global pseudospectral discretisation is well suited to it, and a single Legendre grid of 40 nodes is enough: the maximum relative local error is  $4.0 \times 10^{-9}$  without any mesh refinement. The computed minimum time agrees with the analytic cycloid solution to eight significant figures, which is a more useful check on an installation than any statement in this document.

Table 1: Brachistochrone problem: summary

|                            |                                                                                                                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | The smallest complete problem: single phase, free final time, terminal conditions as event constraints, global pseudospectral discretisation on a fixed mesh                                                    |
| <i>Reference</i>           | Bryson and Ho [4]                                                                                                                                                                                               |
| <i>Dimensions</i>          | 1 phase; 3 states, 1 control, 5 events, no path constraints, no static parameters                                                                                                                               |
| <i>Objective</i>           | $J = t_f$                                                                                                                                                                                                       |
| <i>Dynamics</i>            | $\dot{x} = v \sin \theta, \quad \dot{y} = v \cos \theta, \quad \dot{v} = g \cos \theta$                                                                                                                         |
| <i>Initial conditions</i>  | $x(0) = y(0) = v(0) = 0, \quad t_0 = 0$                                                                                                                                                                         |
| <i>Terminal conditions</i> | $x(t_f) = 2, \quad y(t_f) = 2$                                                                                                                                                                                  |
| <i>Bounds</i>              | $0 \leq x, y, v \leq 20, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq t_f \leq 10$                                                                                                                               |
| <i>Constants</i>           | $g = 9.8$                                                                                                                                                                                                       |
| <i>Discretisation</i>      | Legendre pseudospectral, 40 nodes, automatic differentiation, automatic scaling, NLP tolerance $10^{-6}$                                                                                                        |
| <i>Mesh refinement</i>     | None (fixed mesh)                                                                                                                                                                                               |
| <i>Known solution</i>      | The cycloid $x = k(\phi - \sin \phi), y = k(1 - \cos \phi), t = \sqrt{k/g} \phi$ . The terminal condition $x(t_f) = y(t_f)$ gives $\phi_f = 2.412011143914$ , $k = 1.145834075064$ and $t_f^* = 0.824759143157$ |
| <i>Computed</i>            | $t_f = 8.247591 \times 10^{-1}$ , a relative error of $5 \times 10^{-8}$ against $t_f^*$                                                                                                                        |
| <i>Source</i>              | <code>examples/brac1/brac1.cxx</code> , listed below                                                                                                                                                            |

## Results

### PSOPT results summary

=====

Problem: Brachistochrone Problem

CPU time (seconds): 6.461530e-01

NLP solver used: IPOPT

Optimal (unscaled) cost function value: 8.247591e-01

Phase 1 endpoint cost function value: 8.247591e-01

Phase 1 integrated part of the cost: 0.000000e+00

Phase 1 initial time: 0.000000e+00

```

Phase 1 final time: 8.247591e-01
Phase 1 maximum relative local error: 4.040344e-09
NLP solver reports: The problem has been solved!

```

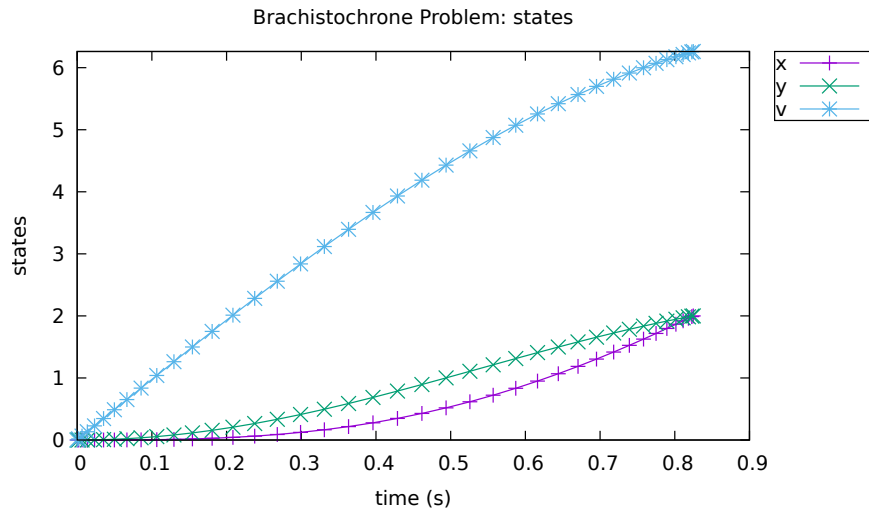


Figure 1: States for the brachistochrone problem

## Source

The complete source of this example is listed below. It is worth reading once in full, because every later example has the same skeleton: the user functions `endpoint_cost`, `integrand_cost`, `dae`, `events` and `linkages` (with `observation_function` added for estimation problems), then the two-level problem setup, the bounds, the initial guess, the algorithm options, and the call to `psopt`.

```

/////////////////////////////////////////////////////////////////
/////////////////////////////////////////////////////////////////
//                               brac1.cxx                               //
/////////////////////////////////////////////////////////////////
//                               PSOPT Example                          //
/////////////////////////////////////////////////////////////////
// Title:           Brachistochrone problem                          //
// Last modified:    04 January 2009                                //
// Reference:        Bryson and Ho (1975)                           //
// (See PSOPT handbook for full reference)                          //
// Copyright (c) Victor M. Becerra, 2009                            //
// This is part of the PSOPT software library, which               //
// is distributed under the terms of the GNU Lesser                 //
// General Public License (LGPL)                                    //
/////////////////////////////////////////////////////////////////

#include "psopt.h"

```

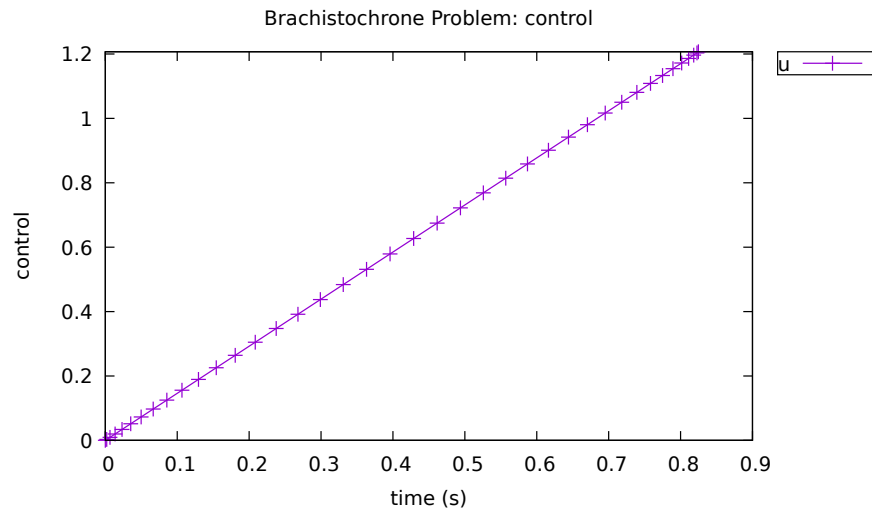


Figure 2: Control for the brachistochrone problem

```
using namespace std;
using namespace PSOPT;

////////////////////////////////////
//////////////////////////////////// Define the end point (Mayer) cost function ///////////////////////////////////
////////////////////////////////////

adouble endpoint_cost(adouble* initial_states, adouble* final_states,
                      adouble* parameters, adouble& t0, adouble& tf,
                      adouble* xad, int iphase, Workspace* workspace)
{
    return tf;
}

////////////////////////////////////
//////////////////////////////////// Define the integrand (Lagrange) cost function ///////////////////////////////////
////////////////////////////////////

adouble integrand_cost(adouble* states, adouble* controls, adouble* parameters,
                      adouble& time, adouble* xad, int iphase, Workspace* workspace)
{
    return 0.0;
}

////////////////////////////////////
//////////////////////////////////// Define the DAE's ///////////////////////////////////
////////////////////////////////////

void dae(adouble* derivatives, adouble* path, adouble* states,
         adouble* controls, adouble* parameters, adouble& time,
         adouble* xad, int iphase, Workspace* workspace)
{
    adouble xdot, ydot, vdot;

    adouble x = states[ 0 ];
    adouble y = states[ 1 ];
    adouble v = states[ 2 ];

    adouble theta = controls[ 0 ];

    xdot = v*sin(theta);
    ydot = v*cos(theta);
    vdot = 9.8*cos(theta);

    derivatives[ 0 ] = xdot;
}
```

```

    derivatives[ 1 ] = ydot;
    derivatives[ 2 ] = vdot;
}

////////////////////////////////////
//////////////////////////////////// Define the events function ///////////////////////////////////
////////////////////////////////////

void events(adouble* e, adouble* initial_states, adouble* final_states,
           adouble* parameters, adouble& t0, adouble& tf, adouble* xad,
           int iphase, Workspace* workspace)
{
    adouble x0 = initial_states[ 0 ];
    adouble y0 = initial_states[ 1 ];
    adouble v0 = initial_states[ 2 ];
    adouble xf = final_states[ 0 ];
    adouble yf = final_states[ 1 ];

    e[ 0 ] = x0;
    e[ 1 ] = y0;
    e[ 2 ] = v0;
    e[ 3 ] = xf;
    e[ 4 ] = yf;
}

////////////////////////////////////
//////////////////////////////////// Define the phase linkages function ///////////////////////////////////
////////////////////////////////////

void linkages( adouble* linkages, adouble* xad, Workspace* workspace)
{
    // No linkages as this is a single phase problem
}

////////////////////////////////////
//////////////////////////////////// Define the main routine ///////////////////////////////////
////////////////////////////////////

int main(void)
{
    /////////////////////////////////// Declare key structures ///////////////////////////////////
    ///////////////////////////////////

    Alg  algorithm;
    Sol  solution;
    Prob problem;

    /////////////////////////////////// Register problem name ///////////////////////////////////
    ///////////////////////////////////

    problem.name      = "Brachistochrone Problem";

    problem.outfilename = "brac1.txt";

    /////////////////////////////////// Define problem level constants & do level 1 setup ///////////////////////////////////
    ///////////////////////////////////

    problem.nphases    = 1;
    problem.nlinkages   = 0;

    psopt_level1_setup(problem);

    /////////////////////////////////// Define phase related information & do level 2 setup ///////////////////////////////////
    ///////////////////////////////////

    problem.phases(1).nstates = 3;
    problem.phases(1).ncontrols = 1;
    problem.phases(1).nevents = 5;
    problem.phases(1).npath = 0;
    problem.phases(1).nodes << 40;

    psopt_level2_setup(problem, algorithm);
}

```



```

////////////////////////////////////
//////////////////////////////////// Enter problem bounds information //////////////////////////////////////
////////////////////////////////////

problem.phases(1).bounds.lower.states << 0, 0, 0;
problem.phases(1).bounds.upper.states << 20, 20, 20;

problem.phases(1).bounds.lower.controls << 0.0;
problem.phases(1).bounds.upper.controls << 2*pi;

problem.phases(1).bounds.lower.events << 0, 0, 0, 2, 2;
problem.phases(1).bounds.upper.events << 0, 0, 0, 2, 2;

problem.phases(1).bounds.lower.StartTime = 0.0;
problem.phases(1).bounds.upper.StartTime = 0.0;

problem.phases(1).bounds.lower.EndTime = 0.0;
problem.phases(1).bounds.upper.EndTime = 10.0;

////////////////////////////////////
//////////////////////////////////// Register problem functions //////////////////////////////////////
////////////////////////////////////

problem.integrand_cost = &integrand_cost;
problem.endpoint_cost = &endpoint_cost;
problem.dae = &dae;
problem.events = &events;
problem.linkages = &linkages;

////////////////////////////////////
//////////////////////////////////// set scaling factors (optional) //////////////////////////////////////
////////////////////////////////////

// problem.phases(1).scale.controls = 1.0*ones(1,1);
// problem.phases(1).scale.states = 1.0*ones(3,1);
// problem.phases(1).scale.events = 1.0*ones(5,1);
// problem.phases(1).scale.defects = 1.0*ones(3,1);
// problem.phases(1).scale.time = 1.0;

// problem.scale.objective = 1.0;

////////////////////////////////////
//////////////////////////////////// Define & register initial guess //////////////////////////////////////
////////////////////////////////////

MatrixXd x0(3,20);

x0.row(0) = linspace(0.0,1.0, 20);
x0.row(1) = linspace(0.0,1.0, 20);
x0.row(2) = linspace(0.0,1.0, 20);

problem.phases(1).guess.controls = ones(1,20);
problem.phases(1).guess.states = x0;
problem.phases(1).guess.time = linspace(0.0, 2.0, 20);

////////////////////////////////////
//////////////////////////////////// Enter algorithm options //////////////////////////////////////
////////////////////////////////////

algorithm.nlp_method = "IPOPT";
algorithm.scaling = "automatic";
algorithm.derivatives = "automatic";
algorithm.nlp_iter_max = 1000;
algorithm.nlp_tolerance = 1.e-6;
// algorithm.hessian = "exact";
algorithm.collocation_method = "Legendre";
// algorithm.mesh_refinement = "automatic";

////////////////////////////////////
//////////////////////////////////// Now call PSOPT to solve the problem //////////////////////////////////////

```

```

////////////////////////////////////
    psopt(solution, problem, algorithm);

    if (solution.error_flag) exit(0);

////////////////////////////////////
//////////////////////////////////// Extract relevant variables from solution structure ///////////////////////////////////
////////////////////////////////////

    MatrixXd x = solution.get_states_in_phase(1);
    MatrixXd u = solution.get_controls_in_phase(1);
    MatrixXd t = solution.get_time_in_phase(1);
    MatrixXd H = solution.get_dual_hamiltonian_in_phase(1);
    MatrixXd lambda = solution.get_dual_costates_in_phase(1);

////////////////////////////////////
//////////////////////////////////// Save solution data to files if desired ///////////////////////////////////
////////////////////////////////////

    Save(x,"x.dat");
    Save(u,"u.dat");
    Save(t,"t.dat");
    Save(lambda,"p.dat");

////////////////////////////////////
//////////////////////////////////// Plot some results if desired (requires gnuplot) ///////////////////////////////////
////////////////////////////////////

    plot(t,x,problem.name + ": states", "time (s)", "states", "x y v");

    plot(t,u,problem.name + ": control", "time (s)", "control", "u");

    plot(t,H,problem.name + ": Hamiltonian", "time (s)", "H", "H");

    plot(t,lambda,problem.name + ": costates", "time (s)", "lambda", "lambda_1 lambda_2 lambda_3");

    plot(t,x,problem.name + ": states", "time (s)", "states", "x y v",
         "pdf", "brac1_states.pdf");

    plot(t,u,problem.name + ": control", "time (s)", "control", "u",
         "pdf", "brac1_control.pdf");

    plot(t,H,problem.name + ": Hamiltonian", "time (s)", "H", "H",
         "pdf", "brac1_hamiltonian.pdf");

    plot(t,lambda,problem.name + ": costates", "time (s)", "lambda", "lambda_1 lambda_2 lambda_3",
         "pdf", "brac1_costates.pdf");

}

////////////////////////////////////
//////////////////////////////////// END OF FILE ///////////////////////////////////
////////////////////////////////////

```

### 3 Two-phase Schwartz problem: linking phases

A problem is split into phases when something about it changes at a known instant: a stage separates, a constraint switches on, a model is replaced. *PSOPT* treats each phase as a separate optimal control problem with its own states, controls, mesh and constraints, and joins them with *linkage constraints*, which the user writes in the `linkages` function. This example, due to Schwartz and used as a test problem by Rutquist and Edvall [15], is the smallest one that shows the mechanism working.

The dynamics are the same in both phases, a damped double integrator whose damp-

ing grows with position, and the split is made at  $t = 1$  because the constraints, not the model, change there. In the first phase the state must stay outside an ellipse in the  $(x_1, x_2)$  plane and the control is limited to  $|u| \leq 1$ ; in the second the ellipse is gone and the control limit is relaxed. The cost is a terminal penalty on the distance of the state from the origin at  $t = 2.9$ , so the problem is to get to the origin while avoiding the ellipse on the way.

The three linkage constraints are supplied by the helper `auto_link`, which imposes continuity of every state and of time across the junction: it is the common case, and writing it out by hand is only necessary when the states of the two phases differ in number or in meaning, as in a launch vehicle that sheds a stage.

Two things in the results deserve comment. The terminal cost comes out at  $3.1 \times 10^{-16}$ , that is, the origin is reached to within the tolerance of the NLP solver, which is what the published solution reports. But the maximum relative local error in the second phase is  $1.3 \times 10^{-4}$ , five orders of magnitude larger than in the brachistochrone, and this is not a defect of the solver: with the control limit relaxed to  $\pm 50$  the optimal control has a sharp transient at the start of the phase, reaching about  $-6.9$  within the first few nodes, and a fixed mesh of 60 nodes cannot resolve it. This is exactly the situation that automatic mesh refinement exists for; the Manutec example of Section 14 and the minimum time to climb of Section 15 show what it does with a problem whose solution is hard on a fixed grid.

## Results

```
PSOPT results summary
=====

Problem:  Two phase Schwartz problem
CPU time (seconds): 6.194100e-02
NLP solver used:  IPOPT

Optimal (unscaled) cost function value:  3.138721e-16
Phase 1 endpoint cost function value: 0.000000e+00
Phase 1 integrated part of the cost:  0.000000e+00
Phase 1 initial time: 0.000000e+00
Phase 1 final time: 1.000000e+00
Phase 1 maximum relative local error: 2.654204e-06
Phase 2 endpoint cost function value: 3.138721e-16
Phase 2 integrated part of the cost:  0.000000e+00
Phase 2 initial time: 1.000000e+00
Phase 2 final time: 2.900000e+00
Phase 2 maximum relative local error: 1.279858e-04
NLP solver reports:  The problem has been solved!
```

Table 2: Two-phase Schwartz problem: summary

|                                  |                                                                                                                                                              |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>              | Multiple phases, linkage constraints via <code>auto_link</code> , a nonlinear path constraint active in one phase only, different bounds in different phases |
| <i>Reference</i>                 | Rutquist and Edvall [15]                                                                                                                                     |
| <i>Dimensions</i>                | 2 phases, 3 linkages; phase 1: 2 states, 1 control, 2 events, 1 path constraint; phase 2: 2 states, 1 control, no events, no path constraints                |
| <i>Objective</i>                 | $J = 5 (x_1(t_f)^2 + x_2(t_f)^2)$ , evaluated at the end of phase 2                                                                                          |
| <i>Dynamics</i> (both phases)    | $\dot{x}_1 = x_2, \quad \dot{x}_2 = u - 0.1 (1 + 2x_1^2) x_2$                                                                                                |
| <i>Initial conditions</i>        | $x_1(0) = x_2(0) = 1$                                                                                                                                        |
| <i>Terminal conditions</i>       | None; the terminal state is penalised, not constrained                                                                                                       |
| <i>Path constraint (phase 1)</i> | $1 - 9(x_1 - 1)^2 - \left( \frac{x_2 - 0.4}{0.3} \right)^2 \leq 0$                                                                                           |
| <i>Linkages</i>                  | Continuity of both states and of time at $t = 1$                                                                                                             |
| <i>Bounds</i>                    | Phase 1: $-0.8 \leq x_2,  u  \leq 1$ ; phase 2: $-10 \leq x_2,  u  \leq 50$ ; both: $-20 \leq x_1 \leq 10, x_2 \leq 10$                                      |
| <i>Phase intervals</i>           | $t \in [0, 1]$ and $t \in [1, 2.9]$ , both fixed                                                                                                             |
| <i>Discretisation</i>            | Hermite-Simpson, 60 nodes per phase, automatic differentiation, automatic scaling, NLP tolerance $10^{-6}$                                                   |
| <i>Mesh refinement</i>           | None (fixed mesh)                                                                                                                                            |
| <i>Known solution</i>            | The origin is reachable, so $J^* = 0$                                                                                                                        |
| <i>Computed</i>                  | $J = 3.1 \times 10^{-16}$ ; maximum relative local error $2.7 \times 10^{-6}$ in phase 1 and $1.3 \times 10^{-4}$ in phase 2                                 |
| <i>Source</i>                    | <code>examples/twophase_schwartz/twophase_schwartz.cxx</code> , listed below                                                                                 |

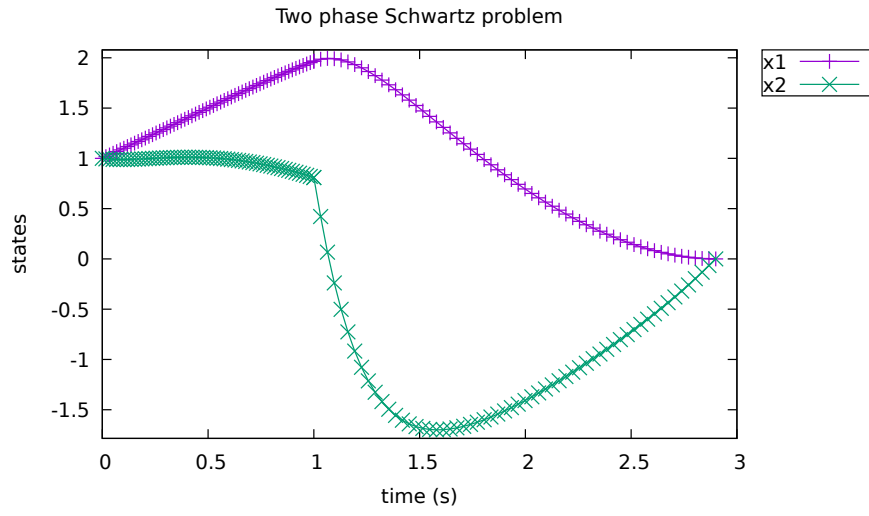


Figure 3: States for the two-phase Schwartz problem; the junction between the phases is at  $t = 1$

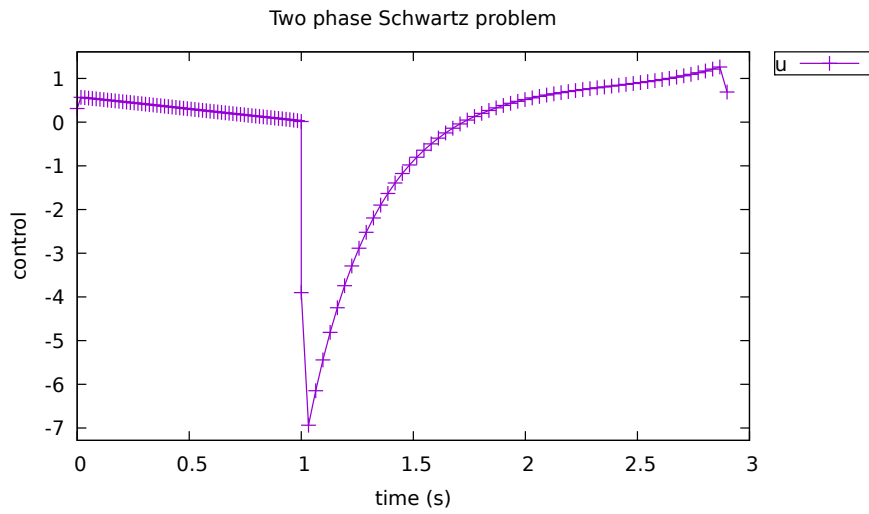


Figure 4: Control for the two-phase Schwartz problem

## Source

The listing below shows how a two-phase problem differs from a single-phase one: `problem.nphases` and `problem.nlinkages` are set before `psopt_level1_setup`, the dimensions and bounds are given per phase, the user functions receive an `iphase` argument and branch on it, and the `linkages` function joins the phases.

```
/////////////////////////////////////////////////////////////////
/////////////////////////////////////////////////////////////////
// twophase_schwartz.cxx //
/////////////////////////////////////////////////////////////////
// PSOPT Example //
/////////////////////////////////////////////////////////////////
// Title: Two phase Schwartz problem //
// Last modified: 09 January 2009 //
// Reference: PROPT Users Guide //
// (See PSOPT handbook for full reference) //
// Copyright (c) Victor M. Becerra, 2009 //
// This is part of the PSOPT software library, which //
// is distributed under the terms of the GNU Lesser //
// General Public License (LGPL) //
/////////////////////////////////////////////////////////////////

#include "psopt.h"

using namespace std;

/////////////////////////////////////////////////////////////////
// Define the end point (Mayer) cost function //
/////////////////////////////////////////////////////////////////

adouble endpoint_cost(adouble* initial_states, adouble* final_states,
                    adouble* parameters, adouble& t0, adouble& tf,
                    adouble* xad, int iphase, Workspace* workspace)
{
    adouble retval;
    adouble x1f, x2f;

    // Phase 1 cost

    if ( iphase == 1 )
    {
        retval = 0.0;
    }

    // Phase 2 cost

    if( iphase == 2 ) {

        x1f = final_states[0];
        x2f = final_states[1];

        retval = 5*(x1f*x1f + x2f*x2f);
    }

    return retval;
}

/////////////////////////////////////////////////////////////////
// Define the integrand (Lagrange) cost function //
/////////////////////////////////////////////////////////////////

adouble integrand_cost(adouble* states, adouble* controls,
                    adouble* parameters, adouble& time, adouble* xad,
                    int iphase, Workspace* workspace)
{
    return 0.0;
}
```

```

////////////////////////////////////
//////////////////////////////////// Define the DAE's //////////////////////////////////
////////////////////////////////////

void dae(adouble* derivatives, adouble* path, adouble* states,
         adouble* controls, adouble* parameters, adouble& time,
         adouble* xad, int iphase, Workspace* workspace)
{
    adouble x1 = states[0];
    adouble x2 = states[1];

    adouble u = controls[0];

    derivatives[0] = x2;
    derivatives[1] = u-0.1*(1+2*x1*x1)*x2;

    if(iphase==1) {
        path[0] = 1.0 - 9.0*((x1-1.0)*(x1-1.0))-((x2-0.4)*(x2-0.4))/(0.3*0.3);
    }
}

////////////////////////////////////
//////////////////////////////////// Define the events function //////////////////////////////////
////////////////////////////////////

void events(adouble* e, adouble* initial_states, adouble* final_states,
           adouble* parameters, adouble& t0, adouble& tf, adouble* xad,
           int iphase, Workspace* workspace)
{
    adouble x1i = initial_states[0];
    adouble x2i = initial_states[1];

    if (iphase==1) {
        e[0] = x1i;
        e[1] = x2i;
    }
}

////////////////////////////////////
//////////////////////////////////// Define the phase linkages function //////////////////////////////////
////////////////////////////////////

void linkages( adouble* linkages, adouble* xad, Workspace* workspace)
{
    int index = 0;
    auto_link( linkages, &index, xad, 1, 2, workspace );
}

////////////////////////////////////
//////////////////////////////////// Define the main routine //////////////////////////////////
////////////////////////////////////

int main(void)
{
    //////////////////////////////////
    ////////////////////////////////// Declare key structures //////////////////////////////////
    //////////////////////////////////

    Alg algorithm;
    Sol solution;
    Prob problem;

    //////////////////////////////////
    ////////////////////////////////// Register problem name //////////////////////////////////
    //////////////////////////////////

    problem.name = "Two phase Schwartz problem";
    problem.outfilename = "twophsc.txt";

    //////////////////////////////////
    ////////////////////////////////// Define problem level constants & do level 1 setup //////////////////////////////////
    //////////////////////////////////

```

```

////////////////////////////////////
problem.nphases    = 2;
problem.nlinkages  = 3;

psopt_level1_setup(problem);

////////////////////////////////////
//////// Define phase related information & do level 2 setup //////////
////////////////////////////////////

problem.phases(1).nstates    = 2;
problem.phases(1).ncontrols  = 1;
problem.phases(1).nevents    = 2;
problem.phases(1).npath     = 1;

problem.phases(2).nstates    = 2;
problem.phases(2).ncontrols  = 1;
problem.phases(2).nevents    = 0;
problem.phases(2).npath     = 0;

problem.phases(1).nodes      << 60;
problem.phases(2).nodes      << 60;

psopt_level2_setup(problem, algorithm);

////////////////////////////////////
//////// Enter problem bounds information //////////
////////////////////////////////////

double x1L = -20.0;
double x2L_phase1 = -0.8;
double x2L_phase2 = -10.0;
double uL  = -1.0;
double x1U = 10.0;
double x2U = 10.0;
double uU  = 1.0;
double hL  = -100.0;
double hU  = 0.0;

// Phase 0 bounds

problem.phases(1).bounds.lower.states(0) = x1L;
problem.phases(1).bounds.lower.states(1) = x2L_phase1;

problem.phases(1).bounds.upper.states(0) = x1U;
problem.phases(1).bounds.upper.states(1) = x2U;

problem.phases(1).bounds.lower.controls(0) = uL;
problem.phases(1).bounds.upper.controls(0) = uU;

problem.phases(1).bounds.lower.events(0) = 1.0;
problem.phases(1).bounds.lower.events(1) = 1.0;

problem.phases(1).bounds.upper.events(0) = 1.0;
problem.phases(1).bounds.upper.events(1) = 1.0;

problem.phases(1).bounds.lower.path(0) = hL;
problem.phases(1).bounds.upper.path(0) = hU;

problem.phases(1).bounds.lower.StartTime    = 0.0;
problem.phases(1).bounds.upper.StartTime    = 0.0;

problem.phases(1).bounds.lower.EndTime      = 1.0;
problem.phases(1).bounds.upper.EndTime      = 1.0;

// Phase 1 bounds

problem.phases(2).bounds.lower.states(0) = x1L;
problem.phases(2).bounds.lower.states(1) = x2L_phase2;

problem.phases(2).bounds.upper.states(0) = x1U;
problem.phases(2).bounds.upper.states(1) = x2U;

problem.phases(2).bounds.lower.controls(0) = -50.0;
problem.phases(2).bounds.upper.controls(0) = 50.0;

problem.phases(2).bounds.lower.StartTime    = 1.0;
problem.phases(2).bounds.upper.StartTime    = 1.0;

```



```

        problem.phases(2).bounds.lower.EndTime      = 2.9;
        problem.phases(2).bounds.upper.EndTime      = 2.9;

////////////////////////////////////
//////////////////////////////////// Register problem functions //////////////////////////////////////
////////////////////////////////////

        problem.integrand_cost = &integrand_cost;
        problem.endpoint_cost = &endpoint_cost;
        problem.dae = &dae;
        problem.events = &events;
        problem.linkages = &linkages;

////////////////////////////////////
//////////////////////////////////// Define & register initial guess //////////////////////////////////////
////////////////////////////////////

        int iphase;

        MatrixXd u0(1,40);
        MatrixXd x0(2,40);

        MatrixXd time_guess0 = linspace(0.0, 1.0 , 40);
        MatrixXd time_guess1 = linspace(1.0, 2.9 , 40);

        iphase = 1;

        u0 = zeros(1,40);
        x0.row(0) = linspace(1.0,1.0, 40);
        x0.row(1) = linspace(1.0,1.0, 40);

        problem.phases(iphase).guess.controls = u0;
        problem.phases(iphase).guess.states = x0;
        problem.phases(iphase).guess.time = time_guess0;

        iphase = 2;

        u0 = zeros(1,40);
        x0.row(0) = linspace(1.0,1.0, 40);
        x0.row(1) = linspace(1.0,1.0, 40);

        problem.phases(iphase).guess.controls = u0;
        problem.phases(iphase).guess.states = x0;
        problem.phases(iphase).guess.time = time_guess1;

////////////////////////////////////
//////////////////////////////////// Enter algorithm options //////////////////////////////////////
////////////////////////////////////

        algorithm.nlp_method          = "IPOPT";
        algorithm.scaling              = "automatic";
        algorithm.derivatives          = "automatic";
        algorithm.collocation_method   = "Hermite-Simpson";
        algorithm.nlp_iter_max         = 1000;
        algorithm.nlp_tolerance        = 1.e-6;

////////////////////////////////////
//////////////////////////////////// Now call PSOPT to solve the problem //////////////////////////////////////
////////////////////////////////////

        psopt(solution, problem, algorithm);

////////////////////////////////////
//////////////////////////////////// Extract relevant variables from solution structure //////////////////////////////////////
////////////////////////////////////

        MatrixXd xphase1 = solution.get_states_in_phase(1);
        MatrixXd uphase1 = solution.get_controls_in_phase(1);
        MatrixXd tphase1 = solution.get_time_in_phase(1);

        MatrixXd xphase2 = solution.get_states_in_phase(2);
        MatrixXd uphase2 = solution.get_controls_in_phase(2);
        MatrixXd tphase2 = solution.get_time_in_phase(2);

        MatrixXd x(xphase1.rows(), xphase1.cols()+xphase2.cols());

```

```

x << xphase1 , xphase2;

MatrixXd u(uptime1.rows(), uptime1.cols()+uptime2.cols());

u << uptime1 , uptime2;

MatrixXd t(uptime1.rows(), uptime1.cols()+uptime2.cols());

t << tphase1 , tphase2;

////////////////////////////////////
////////// Save solution data to files if desired //////////
////////////////////////////////////

Save(x,"x.dat");
Save(u,"u.dat");
Save(t,"t.dat");

////////////////////////////////////
////////// Plot some results if desired (requires gnuplot) //////////
////////////////////////////////////

plot(t,x,problem.name, "time (s)", "states", "x1 x2");

plot(t,u,problem.name, "time (s)", "control", "u");

plot(t,x,problem.name, "time (s)", "states", "x1 x2",
      "pdf", "twophsc_states.pdf");

plot(t,u,problem.name, "time (s)", "control", "u",
      "pdf", "twophsc_control.pdf");

}

////////////////////////////////////
////////// END OF FILE //////////
////////////////////////////////////

```

## 4 Catalytic cracking of gas oil: parameter estimation

Estimating the parameters of a dynamic model from measurements is not a different kind of problem from optimal control: it is an optimal control problem whose decision variables are static parameters rather than a control history, and whose objective is a sum of squared residuals between the model's outputs and the data. *PSOPT* supports it directly. The user declares how many observations there are and at what times they were taken, supplies them, and writes an **observation\_function** that says which combination of states, controls and parameters each measurement corresponds to; the least-squares objective is then formed internally and there is no need to write an integrand or endpoint cost at all.

The problem used here is the catalytic cracking of gas oil, from the COPS collection [5]. Gas oil (species  $A$ ) cracks to gasoline (species  $Q$ ) and to lighter products and coke (species  $S$ ), by the reaction network  $A \rightarrow Q$ ,  $A \rightarrow S$ ,  $Q \rightarrow S$ , in which the first two steps are second order in the concentration of  $A$  and the third is first order in the concentration of  $Q$ . The three rate constants  $\theta_1, \theta_2, \theta_3$  are to be estimated from 21 samples of the two concentrations. The problem is a standard test case because the data are few, the model is nonlinear in the parameters, and the third rate constant is poorly determined, which is the point of interest here.

That last point is what makes the example worth reading rather than merely running. The estimate  $\hat{\theta}_3 \approx 1.0$  arrives with a 95 per cent confidence interval of roughly  $[0.32, 1.69]$ : the data constrain it to within a factor of five, and a report of the estimate alone, without the interval, would be misleading. The correlation matrix explains why. The three parameters are strongly correlated ( $r_{12} = 0.79$ ,  $r_{13} = -0.84$ ,  $r_{23} = -0.87$ ), so the data determine certain combinations of them far better than they determine any one of them; a change in  $\theta_1$  can be compensated almost exactly by changes in  $\theta_2$  and  $\theta_3$ . Setting `algorithm.parameter_statistics = "yes"` makes *PSOPT* compute and report the covariance, the correlations, the residual standard deviation and the confidence limits, and for an estimation problem these should be regarded as part of the answer rather than as optional extras. The statistics are computed by the method of Kostina [8], from the null space of the active constraints at the solution.

The initial conditions  $y(0) = (1, 0)$  are those of the published problem and are imposed here as event constraints. They are not free parameters of the fit: the experiment starts with pure gas oil and no gasoline, and this is known rather than measured.

Table 3: Catalytic cracking of gas oil: summary

|                             |                                                                                                                                                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>         | Parameter estimation: observations, sampling instants, <code>observation_function</code> , and the reporting of covariance, correlations and confidence limits                                     |
| <i>Reference</i>            | Dolan and Moré, COPS [5]; statistics after Kostina [8]                                                                                                                                             |
| <i>Dimensions</i>           | 1 phase; 2 states, no controls, 3 static parameters, 2 events, no path constraints; 2 observed variables at 21 sampling instants                                                                   |
| <i>Objective</i>            | $J = \sum_{i=1}^{21} [(y_1(t_i) - y_{m,1}(i))^2 + (y_2(t_i) - y_{m,2}(i))^2]$ , formed internally from the observations                                                                            |
| <i>Dynamics</i>             | $\dot{y}_1 = -(\theta_1 + \theta_3)y_1^2$ , $\dot{y}_2 = \theta_1 y_1^2 - \theta_2 y_2$                                                                                                            |
| <i>Observation function</i> | $g(x, u, p, t) = [y_1 \ y_2]^T$ , unit residual weights                                                                                                                                            |
| <i>Initial conditions</i>   | $y_1(0) = 1$ , $y_2(0) = 0$                                                                                                                                                                        |
| <i>Bounds</i>               | $0 \leq y_1, y_2 \leq 2$ , $0 \leq \theta_j \leq 20$ , $t \in [0, 0.95]$ fixed                                                                                                                     |
| <i>Data</i>                 | 21 sampling instants and 42 measurements, listed in the source                                                                                                                                     |
| <i>Discretisation</i>       | Hermite-Simpson, 80 nodes, automatic differentiation, automatic scaling, NLP tolerance $10^{-6}$ , <code>parameter_statistics = "yes"</code>                                                       |
| <i>Mesh refinement</i>      | None (fixed mesh)                                                                                                                                                                                  |
| <i>Known solution</i>       | COPS reports a residual of $5.23659 \times 10^{-3}$ . Solving the same problem outside <i>PSOPT</i> by nonlinear least squares with an accurate ODE solver gives $\theta = (11.847, 8.344, 1.001)$ |
| <i>Computed</i>             | $J = 5.228971 \times 10^{-3}$ , $\hat{\theta} = (11.848, 8.345, 1.001)$ , $\hat{\sigma} = 1.158 \times 10^{-2}$                                                                                    |
| <i>Source</i>               | <code>examples/cracking/cracking.cxx</code> , listed below                                                                                                                                         |

## Results

```
PSOPT results summary
=====

Problem: Catalytic cracking of gas oil
CPU time (seconds): 1.155670e-01
NLP solver used: IPOPT

Optimal (unscaled) cost function value: 5.228971e-03
Phase 1 endpoint cost function value: 5.228971e-03
Phase 1 integrated part of the cost: 0.000000e+00
Phase 1 initial time: 0.000000e+00
Phase 1 final time: 9.500000e-01
Phase 1 maximum relative local error: 4.274288e-06
NLP solver reports: The problem has been solved!
```

The parameter statistics reported by *PSOPT* are reproduced below. The confidence limits are the ones to read first, and the correlation matrix immediately after them: the diagonal of the covariance on its own would suggest three independently determined parameters, which these are not.

```
***** PHASE 1 PARAMETERS *****

1.184812e+01
8.345197e+00
1.000611e+00
***** PARAMETER STATISTICS *****
>>>> Parameter covariance matrix (all phases)
1.064561e-01 7.886536e-02 -9.611804e-02
7.886536e-02 9.461554e-02 -9.345352e-02
-9.611804e-02 -9.345352e-02 1.219091e-01

>>>> Rank of parameter covariance matrix: 3

>>>> Parameter correlation matrix (all phases)
1.000000e+00 7.858141e-01 -8.437264e-01
7.858141e-01 1.000000e+00 -8.701547e-01
-8.437264e-01 -8.701547e-01 1.000000e+00

>>>> Estimated residual standard deviation (sigma_hat): 1.157913e-02
>>>> Scalar observations (N_s): 42, degrees of freedom of the NLP (n_f): 3
>>>> Sum of squared weighted residuals at the solution: 5.228971e-03

>>> 95 percent statistical confidence limits on estimated parameters
Phase Parameter (Low Confidence Limit) (Value) (High Confidence Limit)
1 1 1.120848e+01 1.184812e+01 1.248775e+01
1 2 7.742185e+00 8.345197e+00 8.948209e+00
1 3 3.161286e-01 1.000611e+00 1.685094e+00
```

## Source

The listing below includes the measurement vectors and the sampling instants, so it also serves as a template for the mechanics of getting data into a *PSOPT* estimation problem:

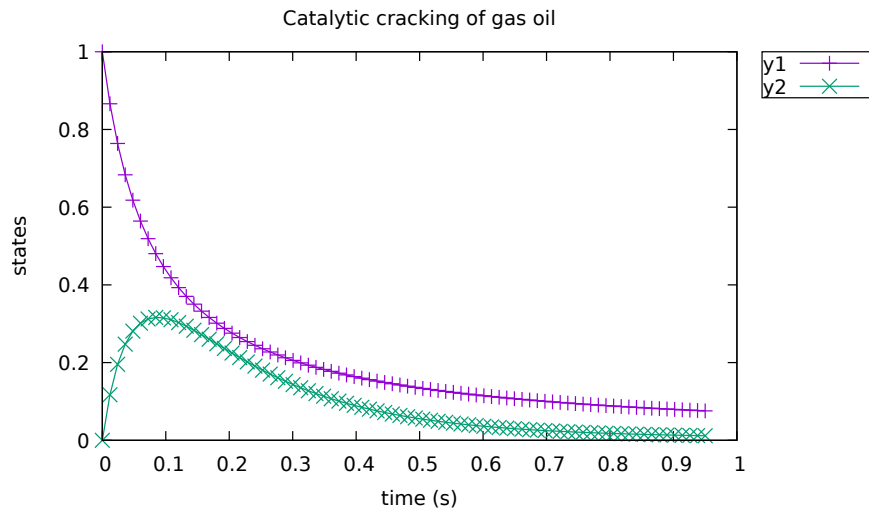


Figure 5: Fitted concentrations and the measurements for the catalytic cracking problem

`nobserved` and `nsamples` are set with the other phase dimensions, `observation_nodes`, `observations` and `residual_weights` are matrices, and `observation_function` is registered alongside the other user functions.

```

////////////////////////////////////
////////////////////////////////////      cracking.cxx      ///////////////////////////////////
////////////////////////////////////
////////////////////////////////////      PSOPT Example      ///////////////////////////////////
////////////////////////////////////
////////////////////////////////////
////////////////////////////////////      Title: Catalytic Cracking of Gas Oil      ///////////////////////////////////
////////////////////////////////////      Last modified: 15 January 2009      ///////////////////////////////////
////////////////////////////////////      Reference: User's guide for DIRCOL      ///////////////////////////////////
////////////////////////////////////      (See PSOPT handbook for full reference)      ///////////////////////////////////
////////////////////////////////////
////////////////////////////////////      Copyright (c) Victor M. Becerra, 2009      ///////////////////////////////////
////////////////////////////////////
////////////////////////////////////      This is part of the PSOPT software library, which      ///////////////////////////////////
////////////////////////////////////      is distributed under the terms of the GNU Lesser      ///////////////////////////////////
////////////////////////////////////      General Public License (LGPL)      ///////////////////////////////////
////////////////////////////////////

#include "psopt.h"

using namespace std;

////////////////////////////////////
////////////////////////////////////      Define the observation function      ///////////////////////////////////
////////////////////////////////////

void observation_function( adouble* observations,
                          adouble* states, adouble* controls,
                          adouble* parameters, adouble& time, int k,
                          adouble* xad, int iphase, Workspace* workspace)
{
    observations[ 0 ] = states[ 0 ];
    observations[ 1 ] = states[ 1 ];
}

```

```

////////////////////////////////////
//////////////////////////////////// Define the DAE's ///////////////////////////////////
////////////////////////////////////

void dae(adouble* derivatives, adouble* path, adouble* states,
         adouble* controls, adouble* parameters, adouble& time,
         adouble* xad, int iphase, Workspace* workspace)
{
    adouble y1 = states[0];
    adouble y2 = states[1];

    adouble theta1 = parameters[ 0 ];
    adouble theta2 = parameters[ 1 ];
    adouble theta3 = parameters[ 2 ];

    derivatives[0] = -(theta1 + theta3)*y1*y1;
    derivatives[1] = theta1*y1*y1 - theta2*y2;
}

////////////////////////////////////
//////////////////////////////////// Define the events function ///////////////////////////////////
////////////////////////////////////

void events(adouble* e, adouble* initial_states, adouble* final_states,
            adouble* parameters, adouble& t0, adouble& tf, adouble* xad,
            int iphase, Workspace* workspace)
{
    // The published problem (COPS gas-oil cracking) fixes y(0) = (1,0);
    // estimating the initial conditions as well makes this a different problem.
    e[0] = initial_states[0];
    e[1] = initial_states[1];
}

////////////////////////////////////
//////////////////////////////////// Define the phase linkages function ///////////////////////////////////
////////////////////////////////////

void linkages( adouble* linkages, adouble* xad, Workspace* workspace)
{
    // No linkages as this is a single phase problem
}

////////////////////////////////////
//////////////////////////////////// Define the main routine ///////////////////////////////////
////////////////////////////////////

int main(void)
{
    MatrixXd y1meas(1,21), y2meas(1,21), tmeas(1,21);
    // Measured values of y1
    y1meas << 1.0,0.8105,0.6208,0.5258,0.4345,0.3903,0.3342,0.3034, \
              0.2735,0.2405,0.2283,0.2071,0.1669,0.153,0.1339,0.1265, \
              0.12,0.099,0.087,0.077,0.069;
    // Measured values of y2
    y2meas << 0.0,0.2,0.2886,0.301,0.3215,0.3123,0.2716,0.2551,0.2258, \
              0.1959,0.1789,0.1457,0.1198,0.0909,0.0719,0.0561,0.046, \
              0.028,0.019,0.014,0.01;
    // Sampling instants
    tmeas << 0.0,0.025,0.05,0.075,0.1,0.125,0.15,0.175,0.2,0.225,0.25, \
             0.3,0.35,0.4,0.45,0.5,0.55,0.65,0.75,0.85,0.95;

    ///////////////////////////////////
    /////////////////////////////////// Declare key structures ///////////////////////////////////
    ///////////////////////////////////

    Alg algorithm;
    Sol solution;
    Prob problem;

    ///////////////////////////////////
    /////////////////////////////////// Register problem name ///////////////////////////////////
    ///////////////////////////////////

    problem.name = "Catalytic cracking of gas oil";
    problem.outfilename = "cracking.txt";
}

```

```

////////////////////////////////////
////////// Define problem level constants & do level 1 setup //////////
////////////////////////////////////

problem.nphases      = 1;
problem.nlinkages     = 0;

psopt_level1_setup(problem);

////////////////////////////////////
////////// Define phase related information & do level 2 setup //////////
////////////////////////////////////

problem.phases(1).nstates   = 2;
problem.phases(1).ncontrols = 0;
problem.phases(1).nevents   = 2;
problem.phases(1).npath     = 0;
problem.phases(1).nparameters = 3;
problem.phases(1).nodes     << 80;
problem.phases(1).nobserved  = 2;
problem.phases(1).nsamples   = 21;

psopt_level2_setup(problem, algorithm);

////////////////////////////////////
////////// Enter estimation information //////////
////////////////////////////////////

MatrixXd observations(2, 21);

observations << y1meas, y2meas;

problem.phases(1).observation_nodes = tmeas;
problem.phases(1).observations      = observations;
problem.phases(1).residual_weights  = ones(2,21);

////////////////////////////////////
////////// Declare DMatrix objects to store results //////////
////////////////////////////////////

DMatrix x, p, t;

////////////////////////////////////
////////// Enter problem bounds information //////////
////////////////////////////////////

problem.phases(1).bounds.lower.states(0) = 0.0;
problem.phases(1).bounds.lower.states(1) = 0.0;

problem.phases(1).bounds.upper.states(0) = 2.0;
problem.phases(1).bounds.upper.states(1) = 2.0;

problem.phases(1).bounds.lower.parameters(0) = 0.0;
problem.phases(1).bounds.lower.parameters(1) = 0.0;
problem.phases(1).bounds.lower.parameters(2) = 0.0;
problem.phases(1).bounds.upper.parameters(0) = 20.0;
problem.phases(1).bounds.upper.parameters(1) = 20.0;
problem.phases(1).bounds.upper.parameters(2) = 20.0;

problem.phases(1).bounds.lower.events(0) = 1.0;
problem.phases(1).bounds.upper.events(0) = 1.0;
problem.phases(1).bounds.lower.events(1) = 0.0;
problem.phases(1).bounds.upper.events(1) = 0.0;

problem.phases(1).bounds.lower.StartTime = 0.0;
problem.phases(1).bounds.upper.StartTime = 0.0;

problem.phases(1).bounds.lower.EndTime = 0.95;
problem.phases(1).bounds.upper.EndTime = 0.95;

////////////////////////////////////
////////// Register problem functions //////////
////////////////////////////////////

problem.dae = &dae;
problem.events = &events;

```

```

problem.linkages = &linkages;
problem.observation_function = & observation_function;

/////////////////////////////////////////////////////////////////
// Define & register initial guess //
/////////////////////////////////////////////////////////////////

MatrixXd state_guess(2, 40);

state_guess.row(0) = linspace(1.0,0.069, 40);
state_guess.row(1) = linspace(0.30,0.01, 40);

problem.phases(1).guess.states      = state_guess;
problem.phases(1).guess.time        = linspace(0.0, 0.95, 40);
problem.phases(1).guess.parameters = zeros(3,1);

/////////////////////////////////////////////////////////////////
// Enter algorithm options //
/////////////////////////////////////////////////////////////////

algorithm.nlp_method      = "IPOPT";
algorithm.scaling          = "automatic";
algorithm.derivatives      = "automatic";
algorithm.collocation_method = "Hermite-Simpson";
algorithm.parameter_statistics = "yes";
algorithm.nlp_iter_max     = 1000;
algorithm.nlp_tolerance    = 1.e-6;
// algorithm.jac_sparsity_ratio = 0.52;

/////////////////////////////////////////////////////////////////
// Now call PSOPT to solve the problem //
/////////////////////////////////////////////////////////////////

psopt(solution, problem, algorithm);

/////////////////////////////////////////////////////////////////
// Extract relevant variables from solution structure //
/////////////////////////////////////////////////////////////////

x = solution.get_states_in_phase(1);
t = solution.get_time_in_phase(1);
p = solution.get_parameters_in_phase(1);

/////////////////////////////////////////////////////////////////
// Save solution data to files if desired //
/////////////////////////////////////////////////////////////////

Save(x,"x.dat");
Save(t,"t.dat");
cout << "\n Estimated parameters\n" << p << endl;
// Print(p,"Estimated parameters");

/////////////////////////////////////////////////////////////////
// Plot some results if desired (requires gnuplot) //
/////////////////////////////////////////////////////////////////

plot(t,x,problem.name, "time (s)", "states", "y1 y2");
plot(t,x,problem.name, "time (s)", "states", "y1 y2",
      "pdf", "cracking_states.pdf");

}

/////////////////////////////////////////////////////////////////
// END OF FILE //
/////////////////////////////////////////////////////////////////

```



## 5 Bryson’s maximum range problem: an equality path constraint

A particle moves in a vertical plane under a constant tangential acceleration  $a$  and gravity  $g$ . Its position is  $(x, y)$ , its speed is  $v$ , and the direction in which it travels is set by a two-component control  $(u_1, u_2)$  that must be a unit vector at every instant. The time available is fixed, the particle starts from rest at the origin, and it must arrive at a prescribed height; subject to that, it is to travel as far as possible in the  $x$  direction. The problem is from Bryson and Ho [4] and it is small enough to be read in a minute, which is why it is used here to introduce the one feature that the brachistochrone of Section 2 did not have: a constraint that must hold along the whole trajectory rather than at its ends.

That constraint is  $u_1^2 + u_2^2 = 1$ . In *PSOPT* a path constraint is a component of the `path` array filled in by the `dae` function, and it is made an equality in exactly the way an event is: by giving its lower and upper bound the same value. Here both are set to unity. Nothing else in the formulation confines the controls in any useful way: their box bounds are  $\pm 10$ , ten times larger than any value they can take. The path constraint is therefore the only thing that gives the two controls their meaning as a steering direction, and it is active at every collocation node by construction. It is worth noticing that the supplied initial guess sets both controls to zero, which violates the constraint outright. That is permitted: the guess has to be a starting point for the NLP solver, not a feasible trajectory, and the interior-point method restores feasibility as it converges.

The solution has no switching structure to look for. Because the right-hand sides do not depend on  $x$  or  $y$ , the costates of those two states are constant, and the stationarity condition with respect to  $(u_1, u_2)$ , taken together with the path constraint, makes the control the unit vector parallel to  $(\lambda_x v, \lambda_y v - g \lambda_v)$ . The computed controls behave accordingly and turn smoothly and monotonically:  $u_2$  begins at  $-1$ , so the particle dives to build up speed, passes through zero shortly after  $t = 1$ , and ends positive as the trajectory is pulled back up to meet  $y(t_f) = 0.1$ ;  $u_1$  rises from zero to unity at the point where the motion is momentarily horizontal, then falls back. The height dips to about  $-0.37$  before recovering and the speed peaks near  $1.37$ . The maximum range obtained is  $x(t_f) = 1.712314$ .

The accuracy deserves a plain word. The run uses trapezoidal collocation on a fixed mesh of 50 nodes with mesh refinement disabled, and the maximum relative local error is  $2.08 \times 10^{-5}$ , four orders of magnitude worse than the pseudospectral result of Section 2. That is expected rather than alarming: the trapezoidal rule is second-order accurate, so its error falls only quadratically with node count, and the NLP tolerance here is  $10^{-4}$  rather than the usual  $10^{-6}$ . Enabling automatic mesh refinement, which the source file leaves commented out, is the remedy if more digits are wanted; the example is kept as it is because it shows what a plain fixed trapezoidal mesh actually delivers.

Table 4: Bryson’s maximum range problem: summary

|                            |                                                                                                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | An equality path constraint, imposed by setting equal lower and upper bounds on a component of the <code>path</code> array; trapezoidal collocation on a fixed mesh        |
| <i>Reference</i>           | Bryson and Ho [4]                                                                                                                                                          |
| <i>Dimensions</i>          | 1 phase; 3 states, 2 controls, 4 events, 1 path constraint, no static parameters                                                                                           |
| <i>Objective</i>           | $J = -x(t_f)$ (maximise the range $x(t_f)$ )                                                                                                                               |
| <i>Dynamics</i>            | $\dot{x} = vu_1, \quad \dot{y} = vu_2, \quad \dot{v} = a - gu_2$                                                                                                           |
| <i>Initial conditions</i>  | $x(0) = y(0) = v(0) = 0, \quad t_0 = 0$                                                                                                                                    |
| <i>Terminal conditions</i> | $y(t_f) = 0.1, \quad t_f = 2$ fixed                                                                                                                                        |
| <i>Path constraints</i>    | $u_1^2 + u_2^2 = 1$ , an equality imposed with coincident bounds                                                                                                           |
| <i>Bounds</i>              | $-10 \leq x, y, v \leq 10, \quad -10 \leq u_1, u_2 \leq 10, \quad t_0 = 0, t_f = 2$                                                                                        |
| <i>Constants</i>           | $g = 1, \quad a = 0.5g$                                                                                                                                                    |
| <i>Discretisation</i>      | Trapezoidal, 50 nodes, automatic differentiation, automatic scaling, state-based defect scaling, limited-memory Hessian, NLP tolerance $10^{-4}$ , ODE tolerance $10^{-6}$ |
| <i>Mesh refinement</i>     | None (fixed mesh)                                                                                                                                                          |
| <i>Known solution</i>      | None quoted. The maximum principle fixes the form of the optimal steering law, but no published numerical value of the maximum range for these data is asserted here       |
| <i>Computed</i>            | $J = -1.712314$ , that is $x(t_f) = 1.712314$ , in 0.06 s of CPU time, with a maximum relative local error of $2.082623 \times 10^{-5}$                                    |
| <i>Source</i>              | <code>examples/bryson_max_range/bryson_max_range.cxx</code>                                                                                                                |

## Results

```
PSOPT results summary
=====
```

```
Problem: Bryson Maximum Range Problem
CPU time (seconds): 5.770000e-02
NLP solver used: IPOPT
```

```
Optimal (unscaled) cost function value: -1.712314e+00
Phase 1 endpoint cost function value: -1.712314e+00
Phase 1 integrated part of the cost: 0.000000e+00
Phase 1 initial time: 0.000000e+00
Phase 1 final time: 2.000000e+00
Phase 1 maximum relative local error: 2.082623e-05
NLP solver reports: The problem has been solved!
```

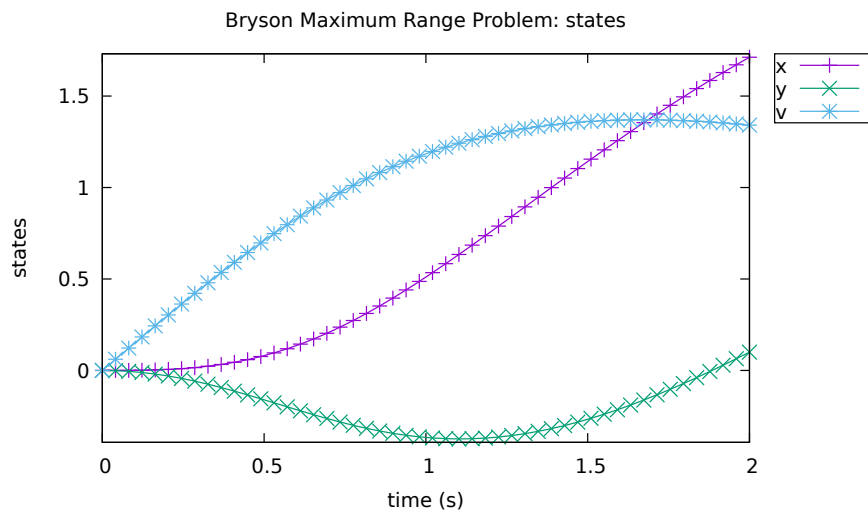


Figure 6: States for Bryson’s maximum range problem

## 6 Interior point constraint: a condition imposed inside the time horizon

Boundary conditions at the two ends of a trajectory are easy to state: they are event constraints, and every example in this document uses them. A condition imposed at an instant *strictly inside* the horizon, such as a spacecraft that must pass through a

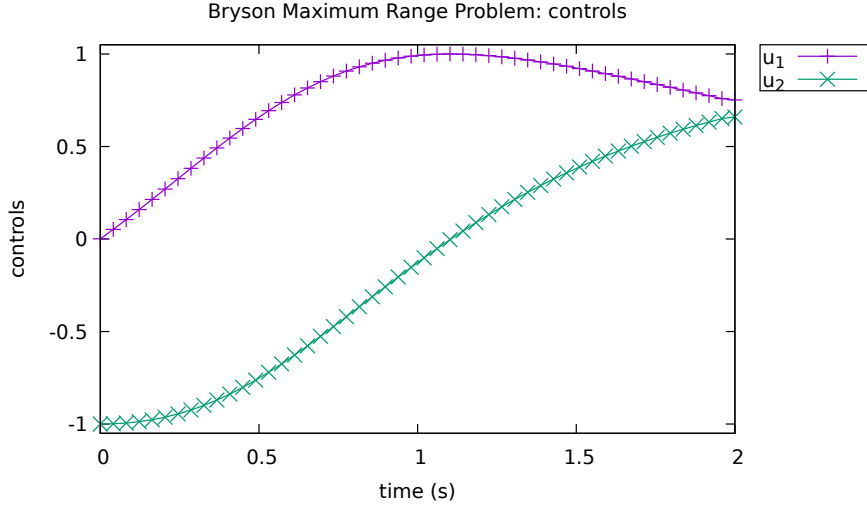


Figure 7: Controls for Bryson’s maximum range problem. The two components trace out a unit vector: the equality path constraint holds along the whole trajectory

given point, a reactor that must reach a given temperature part way through a batch, or a vehicle that must be at a waypoint at a given time, has no such direct expression, because the interface offers events only at  $t_0$  and  $t_f$ . The technique shown here is the standard remedy, and it is the reusable idea of this section: cut the horizon at the instant in question, treat the two pieces as separate phases, and impose the interior condition as an event at the end of the first phase. What was interior to the original problem is terminal to the new first phase, and can therefore be written down.

The problem itself is deliberately trivial so that the construction is visible. It is taken from the MISER3 manual [7]: minimise  $\int_0^1 (x^2 + u^2) dt$  subject to  $\dot{x} = u$  with  $x(0) = 1$  and  $x(1) = 0.75$ , and additionally  $x(0.75) = 0.9$ . The split is at  $t = 0.75$ ; phase 1 runs on  $[0, 0.75]$  and carries two events,  $x(0) = 1$  and  $x(0.75) = 0.9$ , and phase 2 runs on  $[0.75, 1]$  and carries one,  $x(1) = 0.75$ . Both phase boundary times are fixed by giving them equal lower and upper bounds. The two phases are joined by `auto_link`, which generates the two linkage constraints that make the state and the time continuous across the junction; the state continuity is what makes the phase-1 event bind on phase 2 as well, so the interior condition need be written once only.

The splitting is not merely a notational convenience. The optimal control is genuinely discontinuous at  $t = 0.75$ : the Euler-Lagrange equation for this problem is  $\ddot{x} = x$  on each side of the junction, and the two solutions meet with  $u(0.75^-) = 0.2009$  and  $u(0.75^+) = -0.7057$ . A single-phase discretisation would be trying to represent a jump with a polynomial, which is exactly the situation a phase boundary exists to avoid. The interior condition also binds hard: without it the optimal trajectory would pass through  $x(0.75) = 0.7398$ , and forcing it up to 0.9 raises the cost by 0.1453, from 0.7752 to 0.9205. Because  $x$  must return to 0.9 from below, dipping to a minimum of 0.8773 at

$t = 0.523$ , the state is not monotone on phase 1, which is visible in Figure 8.

Being linear-quadratic, the problem has a closed-form answer, and the agreement is exact to every digit *PSOPT* prints:  $9.205314 \times 10^{-1}$  against an analytic 0.9205314414730, with the two phase contributions matching individually as well. Twenty Legendre nodes per phase suffice, the maximum relative local error is below  $2 \times 10^{-9}$  in both phases, no mesh refinement was needed, and the run takes 0.11 s. Neither the state bound  $|x| \leq 1$  nor the control bound  $|u| \leq 1$  is active anywhere. The same construction with the junction time left free, rather than fixed at 0.75, handles the more common case in which the condition must be met at an instant that is itself unknown: a phase boundary is free exactly when its *StartTime* and *EndTime* bounds are given a range rather than a value, as the final phase of the launch problem of Section 16 does.

Table 5: Interior point constraint problem: summary

|                            |                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | Imposing a constraint at an interior instant by splitting the horizon into two phases and writing the constraint as a terminal event of the first phase; automatic phase linkages                                                                                                                                                                      |
| <i>Reference</i>           | MISER3 manual, Jennings et al. [7]                                                                                                                                                                                                                                                                                                                     |
| <i>Dimensions</i>          | 2 phases; 1 state and 1 control per phase, no static parameters; 2 events in phase 1 and 1 in phase 2; 2 linkage constraints; no path constraints                                                                                                                                                                                                      |
| <i>Objective</i>           | $J = \int_0^1 (x^2 + u^2) dt$ , accumulated over both phases                                                                                                                                                                                                                                                                                           |
| <i>Dynamics</i>            | $\dot{x} = u$ in both phases                                                                                                                                                                                                                                                                                                                           |
| <i>Initial conditions</i>  | $x(0) = 1, \quad t_0 = 0$                                                                                                                                                                                                                                                                                                                              |
| <i>Terminal conditions</i> | $x(1) = 0.75, \quad t_f = 1$                                                                                                                                                                                                                                                                                                                           |
| <i>Interior conditions</i> | $x(0.75) = 0.9$ , imposed as the second event of phase 1, whose end time is fixed at 0.75                                                                                                                                                                                                                                                              |
| <i>Linkages</i>            | <code>auto_link</code> between phases 1 and 2: continuity of $x$ and of $t$ at $t = 0.75$                                                                                                                                                                                                                                                              |
| <i>Bounds</i>              | $-1 \leq x \leq 1, \quad -1 \leq u \leq 1$ in both phases; neither is active at the solution                                                                                                                                                                                                                                                           |
| <i>Discretisation</i>      | Legendre pseudospectral, standard differentiation matrix, 20 nodes per phase, automatic differentiation, automatic scaling, limited-memory Hessian, NLP tolerance $10^{-6}$                                                                                                                                                                            |
| <i>Mesh refinement</i>     | None (fixed mesh)                                                                                                                                                                                                                                                                                                                                      |
| <i>Known solution</i>      | The Euler-Lagrange equation is $\ddot{x} = x$ , so on a segment of length $T$ joining $x = \alpha$ to $x = \beta$ the optimal cost is $[(\alpha^2 + \beta^2) \cosh T - 2\alpha\beta] / \sinh T$ . Summing the segments $(1 \rightarrow 0.9, T = 0.75)$ and $(0.9 \rightarrow 0.75, T = 0.25)$ gives $J^* = 0.6607876554 + 0.2597437861 = 0.9205314415$ |
| <i>Computed</i>            | $J = 9.205314 \times 10^{-1}$ , with phase contributions $6.607877 \times 10^{-1}$ and $2.597438 \times 10^{-1}$ : agreement with $J^*$ to all seven figures printed                                                                                                                                                                                   |
| <i>Source</i>              | <code>examples/interior_point/interior_point.cxx</code>                                                                                                                                                                                                                                                                                                |

## Results

```
PSOPT results summary
=====
```

```
Problem: Problem with interior point constraint
CPU time (seconds): 1.096470e-01
NLP solver used: IPOPT
```

```
Optimal (unscaled) cost function value: 9.205314e-01
Phase 1 endpoint cost function value: 0.000000e+00
Phase 1 integrated part of the cost: 6.607877e-01
Phase 1 initial time: 0.000000e+00
Phase 1 final time: 7.500000e-01
Phase 1 maximum relative local error: 1.024202e-09
Phase 2 endpoint cost function value: 0.000000e+00
Phase 2 integrated part of the cost: 2.597438e-01
Phase 2 initial time: 7.500000e-01
Phase 2 final time: 1.000000e+00
Phase 2 maximum relative local error: 2.200546e-10
NLP solver reports: The problem has been solved!
```

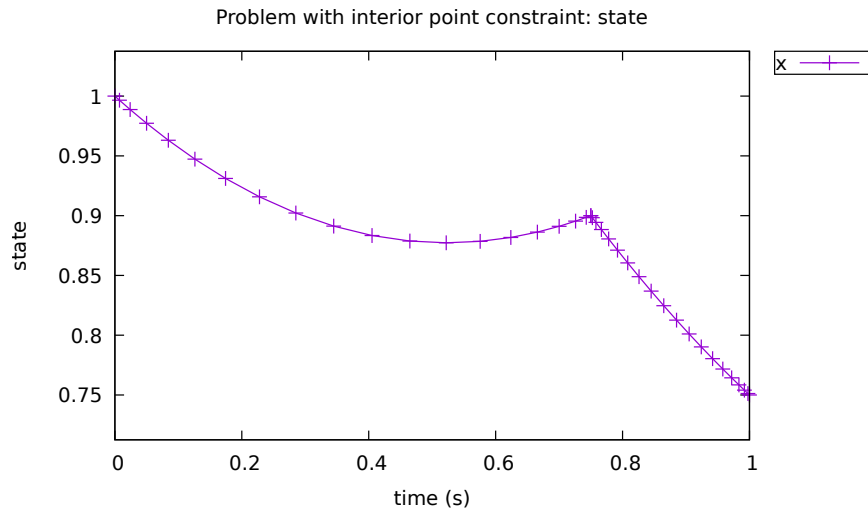


Figure 8: State for the interior point constraint problem. The trajectory is forced through  $x = 0.9$  at  $t = 0.75$  and dips below that value beforehand

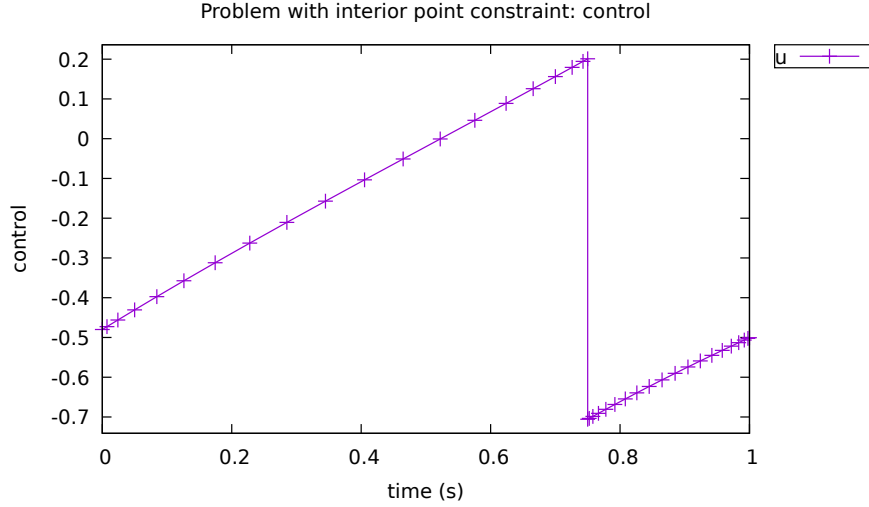


Figure 9: Control for the interior point constraint problem. The jump at  $t = 0.75$  is the phase boundary

## 7 Singular control: a problem whose optimal control is undetermined on part of the interval

A control problem is called singular when the Hamiltonian depends linearly on the control over some interval of time, so that the stationarity condition of the minimum principle degenerates: the coefficient multiplying the control, the switching function, vanishes identically there and  $\partial H/\partial u = 0$  is satisfied for every admissible value of  $u$ . The condition therefore says nothing about what the control should be, and the singular control has to be recovered by differentiating the switching function along the trajectory until the control reappears. Away from such an interval the same problem is bang-bang, with the control resting on one of its bounds. Singular arcs are the standard difficult case for any direct method, which is why one example of the kind is included here.

The example is a third-order problem on the fixed interval  $t \in [0, 1]$ , taken from the singular-control literature [9, 15]. Its Hamiltonian contains the group  $(\lambda_3 - \lambda_2 x_3)u$ , and the only other dependence on the control comes from the quadratic penalty in the integrand, which contributes a quartic term in  $u$  whose coefficient is  $5 \times 10^{-6}x_3^2$ , small enough that over most of the interval the problem behaves as if it were exactly singular. Note that time enters the cost and the second differential equation explicitly, as the independent variable rather than as an extra state: the example carries three states, one control, three events and no path constraints.

Direct collocation responds to this in a characteristic way. The discretised cost is almost insensitive to the control values on a singular portion, so the reduced Hessian of the NLP is nearly singular in those variables and neighbouring nodes can trade control values against one another at almost no penalty. States and objective, which depend on

the control through integrals, come out accurately while the control itself is polluted by node-to-node ringing. The computed control here sits on its upper bound  $u = 10$  until  $t \approx 0.06$ , dips to 7.43 near  $t = 0.14$  and returns to the bound, falls to the lower bound at  $t \approx 0.25$  and stays there until  $t \approx 0.72$ , and then rises to 10 before coming off the bound onto a smooth interior arc that decreases to about 5.19 at  $t \approx 0.94$  and recovers to 6.58 at the final time. The arcs at 10 and at 0 are constrained; the smooth interior arc at the end is the singular one. Two blemishes are visible in Figure 11: a single node at  $t = 0.265$  takes the value 2.28 that neither of its neighbours supports, and along the zero segment the control alternates between about  $2 \times 10^{-6}$  and  $2 \times 10^{-5}$  from one node to the next. Both are artefacts of the flat direction just described, not features of the solution.

The states, by contrast, are clean. The third state is driven through zero at  $t \approx 0.28$  and held to within  $5 \times 10^{-3}$  of it until  $t \approx 0.72$ , which is what makes the coefficient of  $u$  in the second differential equation vanish there, and the trajectory ends at  $x = (-0.1923, -0.0869, 1.7815)$  with a cost of  $1.192620 \times 10^{-1}$ . The maximum relative local error over the single 100-node Legendre mesh is  $7.30 \times 10^{-5}$ , with a mean of  $1.74 \times 10^{-5}$  and a minimum of  $1.23 \times 10^{-7}$ : four orders of magnitude worse than a smooth problem such as the brachistochrone of Section 2, and the discontinuous control is the reason. Note that the estimate is measuring the residual of the transcription's own state and control polynomials, so the ringing on the singular arc, which is a flat direction of the discrete problem rather than an inconsistency with the dynamics, contributes very little to it: the figure is small next to how wrong the control looks. Mesh refinement is switched off in this example, so what is reported is the error of the mesh that was asked for. Refinement would place nodes around the junctions and improve the error there, but it would not by itself remove the ringing on the singular arc, which comes from the structure of the problem rather than from the resolution of the grid.

## Results

```
PSOPT results summary
=====

Problem: Singular control 5
CPU time (seconds): 7.488316e+00
NLP solver used: IPOPT

Optimal (unscaled) cost function value: 1.192620e-01
Phase 1 endpoint cost function value: 0.000000e+00
Phase 1 integrated part of the cost: 1.192620e-01
Phase 1 initial time: 0.000000e+00
Phase 1 final time: 1.000000e+00
Phase 1 maximum relative local error: 7.299454e-05
NLP solver reports: The problem has been solved!
```



Table 6: Singular control problem: summary

|                            |                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | A problem with a singular arc: how a direct collocation method behaves when the optimal control is not determined pointwise by the first-order conditions                         |
| <i>Reference</i>           | Luus [9]; also used as a test problem elsewhere [15]                                                                                                                              |
| <i>Dimensions</i>          | 1 phase; 3 states, 1 control, 3 events, no path constraints, no static parameters                                                                                                 |
| <i>Objective</i>           | $J = \int_0^1 \left[ x_1^2 + x_2^2 + 0.0005 (x_2 + 16t - 8 - 0.1 x_3 u^2)^2 \right] dt$                                                                                           |
| <i>Dynamics</i>            | $\dot{x}_1 = x_2, \quad \dot{x}_2 = -x_3 u + 16t - 8, \quad \dot{x}_3 = u$                                                                                                        |
| <i>Initial conditions</i>  | $x_1(0) = 0, \quad x_2(0) = -1, \quad x_3(0) = -\sqrt{5}$                                                                                                                         |
| <i>Terminal conditions</i> | None; the final state is free                                                                                                                                                     |
| <i>Bounds</i>              | $0 \leq u(t) \leq 10$ ; the states are unbounded; $t_0 = 0$ and $t_f = 1$ are both fixed                                                                                          |
| <i>Discretisation</i>      | Legendre pseudospectral with the standard differentiation matrix, 100 nodes, automatic differentiation, automatic scaling, Jacobian-based defect scaling, NLP tolerance $10^{-7}$ |
| <i>Mesh refinement</i>     | None (fixed mesh)                                                                                                                                                                 |
| <i>Known solution</i>      | No closed-form solution, and no published objective value is quoted here                                                                                                          |
| <i>Computed</i>            | $J = 1.192620 \times 10^{-1}, \quad x(t_f) = (-0.1923, -0.0869, 1.7815),$<br>maximum relative local error $7.30 \times 10^{-5}$                                                   |
| <i>Source</i>              | <code>examples/singular5/singular5.cxx</code>                                                                                                                                     |

The states are shown in Figure 10 and the control in Figure 11. The control figure is the one to look at: the curve joins the computed value at every collocation node, so the node-to-node ringing appears in it directly rather than having to be inferred.

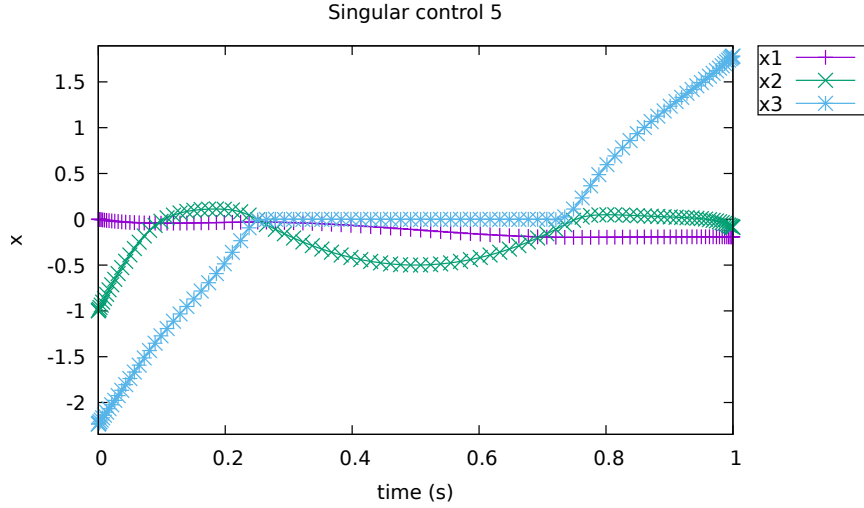


Figure 10: States for the singular control problem

## 8 Time delay problem: states that depend on their own past

A delay differential equation makes the derivative at time  $t$  depend on the state at an earlier time  $t - \tau$ , so the trajectory is no longer determined by its value at a single instant: it needs a history over an interval. Delays of this kind are unavoidable wherever transport, measurement or communication takes time: a reactant travelling down a pipe, a sensor with a filter, a control signal sent over a network. The example here, taken from Luus [9], is a linear second-order system in which both states appear delayed by  $\tau = 0.25$ , driven over a fixed horizon of five seconds to minimise a quadratic cost. The cost is accumulated in a third state so that the objective is a pure endpoint cost, which also makes the running cost readable off the state trajectory.

*PSOPT* evaluates delayed quantities by interpolating the discretised trajectory it already has. The user calls `get_delayed_state` or `get_delayed_control` inside the `dae` function, giving a state or control index and a delay, and receives an `adouble` that participates in the derivative tape like any other. The indices count from zero, as they do everywhere else in the interface; an out-of-range index is caught, but an index that is merely wrong reads the wrong variable silently. For delayed times earlier than  $t_0$  the value has to come from somewhere outside the discretisation. A history function can be registered as `problem.initial_history`; this example does not register one, so

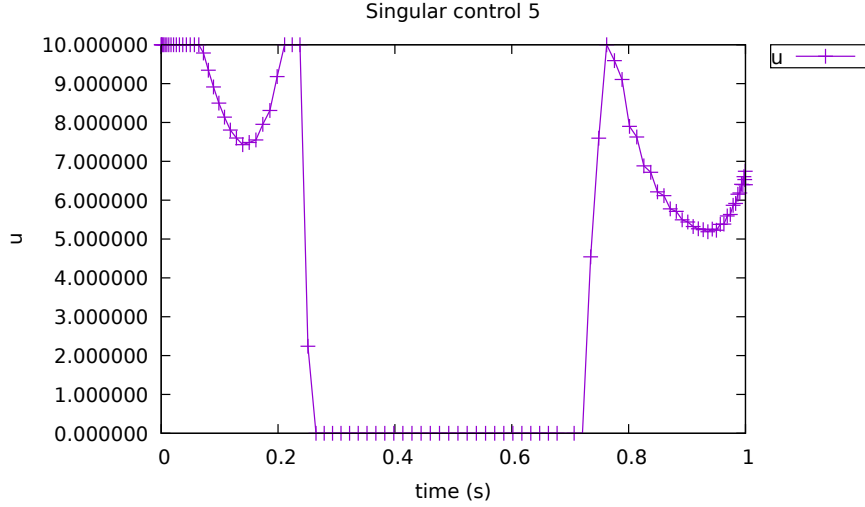


Figure 11: Control for the singular control problem: the arcs at  $u = 10$  and  $u = 0$  lie on the bounds, and the smooth interior arc after  $t \approx 0.78$  is the singular one

*PSOPT* falls back on its default constant history  $\phi(t) = x(t_0)$  and  $\psi(t) = u(t_0)$ . Here that means the delayed terms contribute the fixed amount  $-2x_1(0) - x_2(0) = -3$  over the whole of the first quarter second. The initial time is fixed at zero, which matters more than it looks: the comparison that decides whether a delayed time falls before  $t_0$  is recorded on the tape, so a phase with delays and a free  $t_0$  earns a warning from *PSOPT* and needs numerical derivatives.

The price of the delay is paid in accuracy and in time. The solution has a kink wherever the history discontinuity has propagated forward, that is at every multiple of  $\tau$ , and across such a point a Hermite-Simpson interval behaves like a second-order rather than a fourth-order scheme. The example therefore starts from 60 nodes rather than the 30 a smooth problem of this size would want (a note in the source records that 30 nodes leaves a maximum relative local error of  $1.1 \times 10^{-3}$ ), and leaves the rest to mesh refinement. Two iterations were needed (Table 8): a trapezoidal pass on the initial 60 nodes reaching  $1.37 \times 10^{-3}$ , just short of the  $10^{-3}$  tolerance, and then Hermite-Simpson on 67 nodes at  $3.11 \times 10^{-5}$ . The run takes 37 seconds, which is a great deal for a three-state linear problem, and the reason is visible in the Jacobian: because every delayed value is an interpolation across the trajectory, each collocation constraint touches nodes far from itself and the sparsity ratio is 0.19 at the first iteration, where a local method without delays would give a narrow band.

The discretised problem is a convex quadratic program, since the dynamics of  $x_1$  and  $x_2$  are linear and the cost is quadratic, so it can be solved to high accuracy by other means. An independent solution carried out for this document, on uniform meshes of 200, 400 and 800 intervals, gives 2.529328, 2.526627 and 2.525950, extrapolating to  $J^* = 2.52572$ . The computed value 2.525754 differs from that by about fourteen parts

in a million.

Table 7: Time delay problem: summary

|                            |                                                                                                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | Delayed states in the dynamics: <code>get_delayed_state</code> with zero-based indices, the default constant initial history, and the mesh refinement needed when the history discontinuity propagates                              |
| <i>Reference</i>           | Luus [9]                                                                                                                                                                                                                            |
| <i>Dimensions</i>          | 1 phase; 3 states, 1 control, 3 events, no path constraints, no static parameters                                                                                                                                                   |
| <i>Objective</i>           | $J = x_3(t_f)$                                                                                                                                                                                                                      |
| <i>Dynamics</i>            | $\dot{x}_1 = x_2(t), \quad \dot{x}_2 = -10x_1(t) - 5x_2(t) - 2x_1(t - \tau) - x_2(t - \tau) + u(t),$<br>$\dot{x}_3 = \frac{1}{2} (10x_1^2(t) + x_2^2(t) + u^2(t))$                                                                  |
| <i>Initial conditions</i>  | $x_1(0) = 1, \quad x_2(0) = 1, \quad x_3(0) = 0$ ; history $\phi(t) = x(0)$ for $t < 0$ , the <i>PSOPT</i> default                                                                                                                  |
| <i>Terminal conditions</i> | None; $t_f = 5$ fixed                                                                                                                                                                                                               |
| <i>Bounds</i>              | $-100 \leq x_1, x_2, x_3 \leq 100, \quad -100 \leq u \leq 100, \quad t_0 = 0, t_f = 5$                                                                                                                                              |
| <i>Constants</i>           | $\tau = 0.25$                                                                                                                                                                                                                       |
| <i>Discretisation</i>      | Hermite-Simpson, 60 nodes initially, automatic differentiation, automatic scaling, NLP tolerance $10^{-6}$                                                                                                                          |
| <i>Mesh refinement</i>     | Automatic, local variant, ODE tolerance $10^{-3}$ ; 2 iterations, 60 to 67 nodes, first pass trapezoidal (Table 8)                                                                                                                  |
| <i>Known solution</i>      | No published value is quoted by the source. An independent solution of the same problem for this document, on uniform meshes of 200, 400 and 800 intervals, gives 2.529328, 2.526627 and 2.525950, extrapolating to $J^* = 2.52572$ |
| <i>Computed</i>            | $J = 2.525754$ , a relative difference of $1.4 \times 10^{-5}$ against $J^*$ ; maximum relative local error $3.113100 \times 10^{-5}$                                                                                               |
| <i>Source</i>              | <code>examples/delay1/delay1.cxx</code>                                                                                                                                                                                             |

## Results

### PSOPT results summary

=====

Problem: Time delay problem 1  
CPU time (seconds): 3.676856e+01  
NLP solver used: IPOPT

Optimal (unscaled) cost function value: 2.525754e+00  
Phase 1 endpoint cost function value: 2.525754e+00  
Phase 1 integrated part of the cost: 0.000000e+00  
Phase 1 initial time: 0.000000e+00  
Phase 1 final time: 5.000000e+00

| Iter             | DM  | M  | NV  | NC  | OE | CE | JE | HE | RHS  | $\epsilon_{\max}$ | CPU <sub>a</sub> |
|------------------|-----|----|-----|-----|----|----|----|----|------|-------------------|------------------|
| 1                | TRP | 60 | 242 | 184 | 27 | 27 | 27 | 0  | 3213 | 1.370e-03         | 1.663e+01        |
| 2                | H-S | 67 | 336 | 205 | 11 | 11 | 11 | 0  | 2926 | 3.113e-05         | 1.727e+01        |
| <hr/>            |     |    |     |     |    |    |    |    |      |                   |                  |
| CPU <sub>b</sub> | -   | -  | -   | -   | -  | -  | -  | -  | -    | -                 | 2.873e+00        |
| -                | -   | -  | -   | -   | 38 | 38 | 38 | 0  | 6139 | -                 | 3.677e+01        |

*Key:* Iter=iteration number, DM= discretization method, M=number of nodes, NV=number of variables, NC=number of constraints, OE=objective evaluations, CE = constraint evaluations, JE = Jacobian evaluations, HE = Hessian evaluations, RHS = ODE right hand side evaluations,  $\epsilon_{\max}$  = maximum relative ODE error, CPU<sub>a</sub> = CPU time in seconds spent by NLP algorithm, CPU<sub>b</sub> = additional CPU time in seconds spent by PSOPT

Phase 1 maximum relative local error: 3.113100e-05  
NLP solver reports: The problem has been solved!

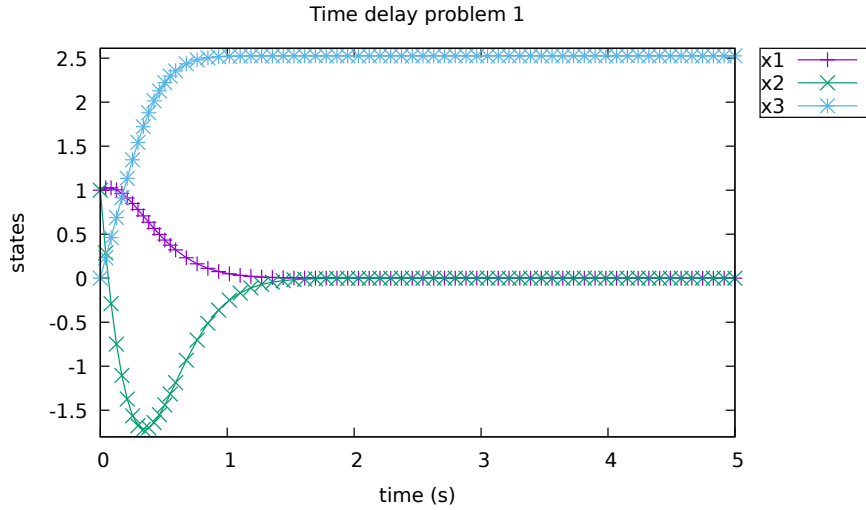


Figure 12: States for the time delay problem

## 9 Constrained pendulum: a high-index DAE, and a second parameter estimation

Write the equations of a point pendulum in Cartesian coordinates and the result is a differential-algebraic equation of index three. The bob is at  $(x_1, x_2)$  with velocity  $(x_3, x_4)$ ; the rod pulls along its own direction with a magnitude not known in advance,

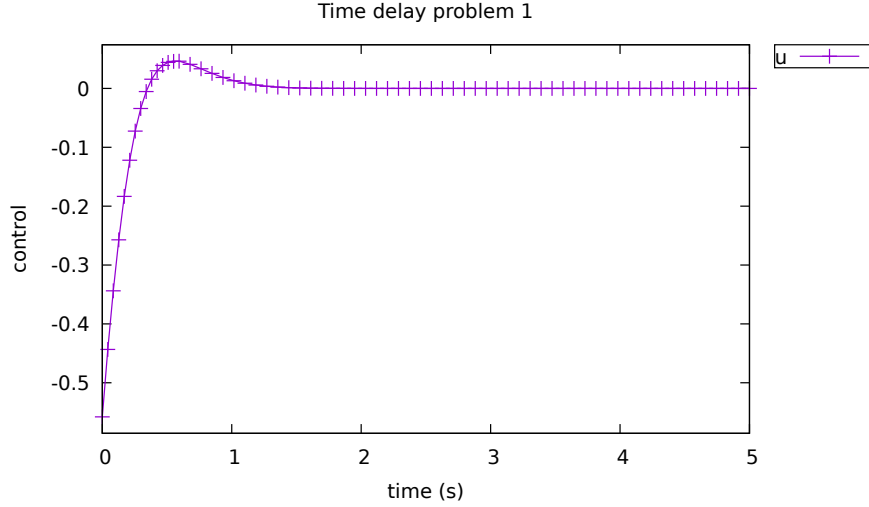


Figure 13: Control for the time delay problem

which enters the equations of motion as a multiplier  $\lambda(t)$ . The differential equations therefore carry an unknown algebraic variable, and the equation that ought to determine it,  $L^2 - x_1^2 - x_2^2 = 0$ , does not contain it. This example takes the version posed by Schittkowski [19], in which  $L$  is not known either and has to be estimated from twenty samples of the two position coordinates, so the section stands both for high-index DAEs and for a second instance of the estimation machinery introduced in Section 4.

The index is the number of times the algebraic equation has to be differentiated before the multiplier appears. Once gives the velocity-level condition  $x_1x_3 + x_2x_4 = 0$ ; twice gives the acceleration level,  $x_3^2 + x_4^2 + \lambda(x_1^2 + x_2^2) = 0$ ; only a third differentiation produces a differential equation for  $\lambda$  itself. That is what makes the system awkward for a great many solvers. Integration-based methods (BDF codes, implicit Runge-Kutta schemes, and any shooting transcription built on them) degrade as the step size falls, because the multiplier is recovered through a divided difference taken three times and its error grows like an inverse power of the step. They also need initial values consistent with the two hidden constraints as well as with the position constraint, an unpleasant demand here, since the initial state is exactly what is not known. The usual remedy, index reduction by substituting a differentiated constraint, buys stability at the cost of drift off the circle  $x_1^2 + x_2^2 = L^2$ , which then has to be stabilised.

*PSOPT* sidesteps all of this in a way worth taking away. The algebraic variable is declared as a control, and the algebraic equation is declared as a path constraint with equal lower and upper bounds. Nothing is integrated: the state polynomial and the algebraic variable are unknowns of the same nonlinear program, and the constraint is imposed as an equality at every collocation node, so it cannot drift. The multiplier is fixed by the algebra of the constraint rather than by differentiating it, and no consistent initialisation is needed: this example declares no event constraints at all, and

the initial state is decided by the data along with everything else. Note also that the two hidden constraints are never written down. Enforcing the position constraint at all thirty nodes of a single global polynomial is enough to imply them to the accuracy of the discretisation.

The fit is essentially exact. On one fixed Legendre grid of 30 nodes the sum of squared residuals is  $2.42 \times 10^{-18}$ , the maximum relative local error is  $9.7 \times 10^{-9}$ , the estimated length is  $\hat{L} = 1.000000$  against the exact value  $L = 1$ , and the computed  $\lambda(t)$  is flat at  $-1$  to the six figures written to the solution file. Mesh refinement was not needed and did not iterate. The parameter statistics should be read with that in mind: the estimated residual standard deviation is  $2.6 \times 10^{-10}$  and the 95 per cent confidence interval on  $\hat{L}$  collapses to the estimate itself when printed to seven figures. That is not evidence that twenty samples fix a pendulum length to ten decimal places; the supplied data are noise-free samples of the analytic solution  $x_1 = \sin(t + \frac{1}{2})$ ,  $x_2 = \cos(t + \frac{1}{2})$ , reproduced to the eight figures held in the data file. At this residual the limits measure the arithmetic rather than the experiment. That is the opposite of the wide intervals in Section 4, and a reminder that an interval is only as informative as the noise behind it.

## Results

```
PSOPT results summary
=====

Problem:   DAE Index 3
CPU time (seconds): 4.478380e-01
NLP solver used: IPOPT

Optimal (unscaled) cost function value: 2.421548e-18
Phase 1 endpoint cost function value: 2.421548e-18
Phase 1 integrated part of the cost: 0.000000e+00
Phase 1 initial time: 5.000000e-01
Phase 1 final time: 1.000000e+01
Phase 1 maximum relative local error: 9.671447e-09
NLP solver reports: The problem has been solved!
```

The parameter statistics are reproduced below. The covariance is  $3.27 \times 10^{-21}$  and the reported degrees of freedom are 40 scalar observations against 3 free directions in the nonlinear program.

```
***** PHASE 1 PARAMETERS *****

1.000000e+00
***** PARAMETER STATISTICS *****
>>>> Parameter covariance matrix (all phases)
3.272362e-21
```

Table 9: Constrained pendulum (DAE index 3): summary

|                             |                                                                                                                                                                                                                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>         | An index-3 differential-algebraic system solved without index reduction: the algebraic state declared as a control and the algebraic equation as an equality path constraint; parameter estimation with a fitted initial state |
| <i>Reference</i>            | Schittkowski [19]; statistics after Kostina [8]                                                                                                                                                                                |
| <i>Dimensions</i>           | 1 phase; 4 differential states, 1 control (the algebraic state $\lambda$ ), 1 static parameter, no events, 1 path constraint; 2 observed variables at 20 sampling instants                                                     |
| <i>Objective</i>            | $J = \sum_{k=1}^{20} [(y_1(t_k) - \hat{y}_1(t_k))^2 + (y_2(t_k) - \hat{y}_2(t_k))^2]$ , formed internally from the observations                                                                                                |
| <i>Dynamics</i>             | $\dot{x}_1 = x_3, \quad \dot{x}_2 = x_4, \quad \dot{x}_3 = \lambda x_1, \quad \dot{x}_4 = \lambda x_2$                                                                                                                         |
| <i>Observation function</i> | $g(x, u, p, t) = [x_1 \ x_2]^T$ , unit residual weights                                                                                                                                                                        |
| <i>Path constraints</i>     | $0 = L^2 - x_1^2 - x_2^2$ , imposed at every node with equal lower and upper bounds                                                                                                                                            |
| <i>Bounds</i>               | $-2 \leq x_j \leq 2, \quad -10 \leq \lambda \leq 10, \quad 0 \leq L \leq 5, \quad t \in [0.5, 10]$ fixed                                                                                                                       |
| <i>Data</i>                 | 20 sampling instants at $t = 0.5, 1.0, \dots, 10.0$ and 40 measurements, read from <code>examples/dae_i3/dae_i3.dat</code>                                                                                                     |
| <i>Discretisation</i>       | Legendre pseudospectral, 30 nodes, standard differentiation matrix, automatic differentiation, automatic scaling, limited-memory Hessian, NLP tolerance $10^{-6}$                                                              |
| <i>Mesh refinement</i>      | None (fixed mesh)                                                                                                                                                                                                              |
| <i>Known solution</i>       | $L = 1$ and $\lambda(t) = -1$ exactly [19]                                                                                                                                                                                     |
| <i>Computed</i>             | $\hat{L} = 1.000000, \quad \lambda(t) = -1$ throughout, $J = 2.421548 \times 10^{-18}$ , $\hat{\sigma} = 2.558266 \times 10^{-10}$ , maximum relative local error $9.671447 \times 10^{-9}$                                    |
| <i>Source</i>               | <code>examples/dae_i3/dae_i3.cxx</code>                                                                                                                                                                                        |



```

>>>> Rank of parameter covariance matrix: 1

>>>> Parameter correlation matrix (all phases)
1.000000e+00

>>>> Estimated residual standard deviation (sigma_hat): 2.558266e-10
>>>> Scalar observations (N_s): 40, degrees of freedom of the NLP (n_f): 3
>>>> Sum of squared weighted residuals at the solution: 2.421548e-18

>>> 95 percent statistical confidence limits on estimated parameters
Phase Parameter (Low Confidence Limit) (Value) (High Confidence Limit)
1 1 1.000000e+00 1.000000e+00 1.000000e+00

```

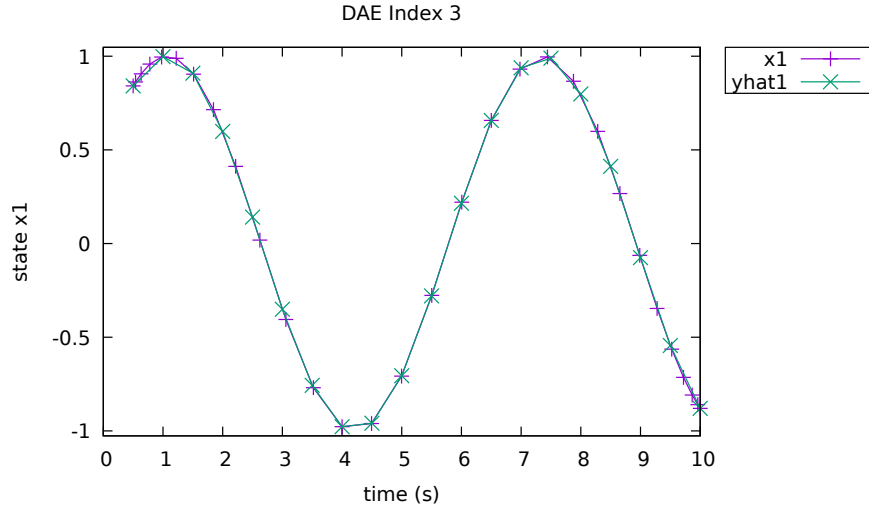


Figure 14: Fitted state  $x_1$  and the measurements for the constrained pendulum

## 10 Integer static parameter: a discrete-valued decision variable

Static parameters that are continuous decision variables appear in several of the problems in this document; the catalytic cracking example (4) is one, where they are the quantities being estimated. Some problems instead have a static parameter that may take only a finite set of values, such as a gear ratio, the number of units in a plant or a discrete valve setting, and rounding whatever the continuous solver returns is not in general the same thing as optimising over the admissible set. *PSOPT* handles the discrete case with a dedicated driver, and the example here is the smallest problem that exercises it, deliberately constructed so that the right answer can be written down by hand.

The system is a double integrator on  $t \in [0, 1]$  whose acceleration is not a control but the static parameter  $p$ :  $\dot{s} = v$ ,  $\dot{v} = p$ , started from rest at the origin, so that  $v(1) = p$  exactly. There is no control variable at all and no integrand cost. The whole problem

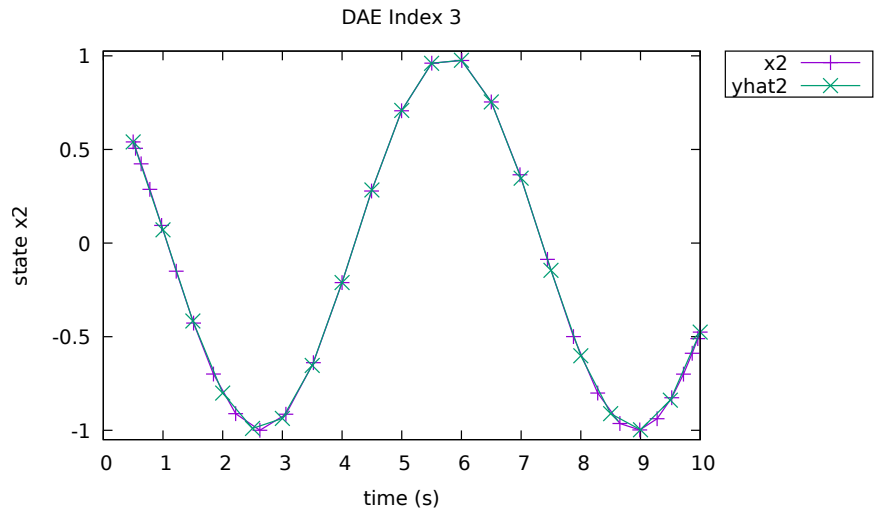


Figure 15: Fitted state  $x_2$  and the measurements for the constrained pendulum

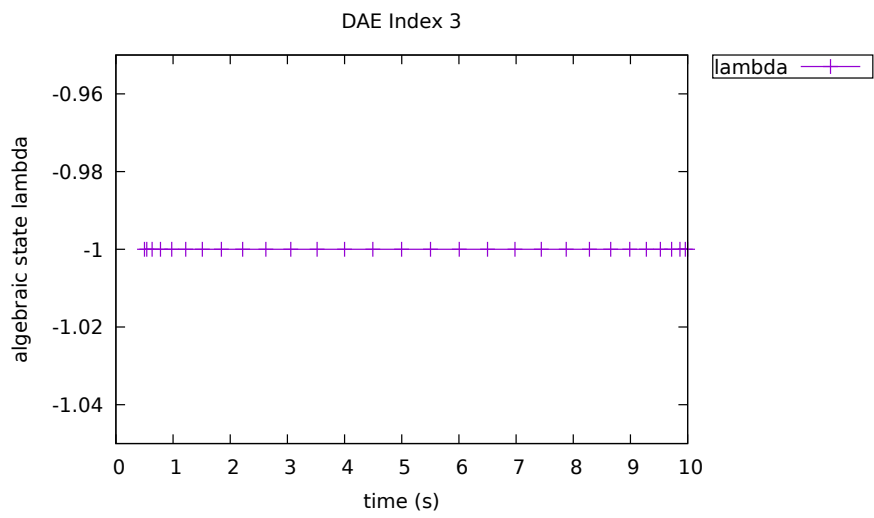


Figure 16: The algebraic state  $\lambda(t)$ , computed as a control

is the Mayer term  $J = (v(1) - c)^2 = (p - c)^2$  with  $c = 2.3$ , and the only decision left is which member of the admissible set  $\{0, 1, 2, 3\}$  the parameter should be. The nearest admissible value to  $c$  is  $p = 2$ , giving  $J = 0.09$ , and the example checks the computed answer against that closed form and prints PASS or FAIL.

The parameter is made discrete by a single call to `declare_integer_parameter`, which takes the problem, the phase, the index of the parameter and the row vector of admissible values; the problem is then handed to `psopt_solve_integer` rather than to `psopt`. That driver enumerates the Cartesian product of every declared admissible set, and for each combination pins the parameter by setting its lower and upper bound to the same value and calling the ordinary `psopt` solver. The best cleanly converged trial is remembered, re-solved once into the caller's `solution`, and the user's original parameter bounds are put back. Five solves therefore take place here: four enumerated trials and the final re-solve. The core solver is untouched by any of this: it sees only a parameter whose bounds happen to coincide. A guard, `algorithm.max_integer_combinations` (default 4096), stops the enumeration before it is attempted if the product of the set sizes is too large, which is the honest statement of where this method stops being usable: it is exact in the small-discrete-space regime and nowhere else.

It is worth being precise about where relaxation and rounding enter, because they do, but not as the mechanism that produces the answer. The bounds the user declares,  $0 \leq p \leq 3$ , are the convex hull of the admissible set; they must contain it, since each enumerated value is imposed inside that box, and they are what the parameter reverts to after the solve. Rounding appears in `reconstruct_integer_parameter`, which reads the parameter back out of the solved phase and snaps it to the nearest admissible value, a safety net against arithmetic noise rather than a decision, since the enumeration has already pinned the value exactly. The naive alternative, solving once over the relaxed box and rounding, would happen to succeed on this problem: the cost is convex in  $p$ , the relaxed minimiser is  $p = 2.3$ , and the nearest admissible value is 2. That coincidence is a property of a scalar convex cost, not a guarantee; enumeration gives the same answer without needing it.

## Results

The example runs with `print_level` set to zero, so the four enumerated solves produce no output of their own and there are no trajectory plots to show: a two-state trajectory determined entirely by a scalar would say nothing that the selected value does not. The report below is the one the example prints itself, and it is the whole of the result.

```
=====
Integer static parameter (target c = 2.300000)
=====

Admissible values : {0, 1, 2, 3}
Solver return code: 0

Selected p      = 2.000000  (admissible index 2)
Optimal cost J  = 0.090000

Closed form     -> p = 2.000000, J = 0.090000
```

Table 10: Integer static parameter: summary

|                           |                                                                                                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>       | A discrete-valued static parameter: <code>declare_integer_parameter</code> , solution by enumeration with <code>psolve_integer</code> , and read-back with <code>reconstruct_integer_parameter</code> |
| <i>Reference</i>          | None. The problem is a constructed analytical test, written for this purpose and not taken from the literature                                                                                        |
| <i>Dimensions</i>         | 1 phase; 2 states, no controls, 1 static parameter (integer, 4 admissible values), 2 events, no path constraints                                                                                      |
| <i>Objective</i>          | $J = (v(t_f) - c)^2 = (p - c)^2$ , $c = 2.3$ (Mayer term only)                                                                                                                                        |
| <i>Dynamics</i>           | $\dot{s} = v$ , $\dot{v} = p$                                                                                                                                                                         |
| <i>Initial conditions</i> | $s(0) = 0$ , $v(0) = 0$                                                                                                                                                                               |
| <i>Path constraints</i>   | None                                                                                                                                                                                                  |
| <i>Bounds</i>             | $p \in \{0, 1, 2, 3\}$ , relaxed box $0 \leq p \leq 3$ ; $-20 \leq s, v \leq 20$ ; $t \in [0, 1]$ fixed                                                                                               |
| <i>Constants</i>          | $c = 2.3$                                                                                                                                                                                             |
| <i>Discretisation</i>     | Legendre pseudospectral (the default), 20 nodes, automatic differentiation, automatic scaling, NLP tolerance $10^{-8}$ , <code>print_level = 0</code> so that the enumerated solves are silent        |
| <i>Mesh refinement</i>    | None (fixed mesh)                                                                                                                                                                                     |
| <i>Known solution</i>     | Closed form: the admissible value nearest $c$ , that is $p^* = 2$ and $J^* = (2 - 2.3)^2 = 0.09$                                                                                                      |
| <i>Computed</i>           | $p = 2.000000$ (admissible index 2), $J = 0.090000$ , solver return code 0; the example's own check passes                                                                                            |
| <i>Source</i>             | <code>examples/int_static_linear/int_static_linear.cxx</code>                                                                                                                                         |

```
=====
Result: PASS
=====
```

## 11 Lotka-Volterra fishing: an integer-valued control

A fishing fleet operates on a prey-predator system and may only be sent out or kept in port: the control is binary. From an initial population pair  $(0.5, 0.7)$  the fleet must be scheduled over a fixed horizon of twelve time units to steer the system to the coexistence steady state  $(1, 1)$ , at least cost in integrated squared deviation from it. This is the Lotka-Volterra fishing problem, the standard small benchmark of mixed-integer optimal control, due to Sager [16] and distributed through mintOC [17]. It stands here for problems in which a decision variable is a switch rather than a dial. Its static counterpart, a parameter drawn from a finite set, is the subject of section 10.

The dynamics are written once, with the fishing effort  $w$  appearing as an ordinary control, and the single call `declare_integer_control` then names that control and its admissible set  $\{0, 1\}$ . Everything else happens inside *PSOPT*. The named control is removed from the control vector and replaced by two weight variables  $\omega_0, \omega_1 \in [0, 1]$ ; the user's right-hand side is evaluated once at  $w = 0$  and once at  $w = 1$  and the two results are combined as  $\omega_0 f(x, 0) + \omega_1 f(x, 1)$ ; and one equality path constraint,  $\omega_0 + \omega_1 = 1$ , is appended. This is outer convexification, and its virtue is that the relaxation it produces is tight in the sense that matters: its optimum is the infimum of the original binary problem. What comes back from the NLP is a fractional  $\omega(t)$ , and `reconstruct_integer_control` turns it into an admissible schedule by sum-up rounding, the accumulator sweep of Sager, Bock and Diehl [18], which activates on each interval whichever mode is most owed relative to its relaxed integral. Their theorem bounds the resulting state deviation by a constant times the mesh spacing  $h$ , so the rounded trajectory, and with it the objective, approaches the relaxed one as  $h \rightarrow 0$ .

The crucial point about this problem is that no optimal integer control exists. The relaxed optimum  $\Phi^* = 1.34408$  reported by mintOC is an infimum that is approached but never attained. The reason is that the relaxed solution is singular over part of the horizon: there  $\omega(t)$  is strictly between zero and one, meaning the fleet is best operated at a fractional rate, and no  $\{0, 1\}$ -valued function equals a fractional one on a set of positive measure. By the approximation theorem the value can be got arbitrarily close by switching fast enough, which is why the switch count in Table 12 climbs with the mesh: 4 switches at 40 nodes, 77 at 640. Each of those schedules is feasible, each better than the last, but the sequence has no best member. This is a property of the problem rather than a defect of the method, and what is reported here is the honest answer: a lower bound from the relaxation, and a sequence of implementable controls descending towards it.

The relaxed objective falls from 1.348102 at 40 nodes to 1.344093 at 640, converging to  $\Phi^*$  from above as the trapezoidal discretisation error shrinks; at the finest mesh the

maximum relative local error is  $2.4 \times 10^{-4}$ . The rounded objective behaves less tidily. It is 1.351850 at 40 nodes but 1.431325 at 80, a gap that is ten times worse on the finer mesh. Nothing has gone wrong: near a singular arc the accumulator can be tipped either way by a small change in the weights, a switch lands one whole interval away from where it should, and the theory offers only a one-sided  $O(h)$  bound, not monotonicity. The fine end is clean, the gap falling from  $5.9 \times 10^{-4}$  at 320 nodes to  $1.3 \times 10^{-4}$  at 640, faster than linearly in  $h$ , which the bound permits.

Table 11: Lotka-Volterra fishing problem: summary

|                           |                                                                                                                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>       | An integer-valued control: <code>declare_integer_control</code> , outer convexification performed internally, and recovery of a binary schedule by sum-up rounding                            |
| <i>Reference</i>          | Sager [16]; mintOC benchmark library [17]; rounding after Sager, Bock and Diehl [18]                                                                                                          |
| <i>Dimensions</i>         | 1 phase; 3 states, 1 control, 3 events, no user path constraints, no static parameters. The declaration expands the phase internally to 2 weight controls and 1 SOS1 equality path constraint |
| <i>Objective</i>          | $J = x_2(t_f) = \int_0^{12} [(x_0 - 1)^2 + (x_1 - 1)^2] dt$                                                                                                                                   |
| <i>Dynamics</i>           | $\dot{x}_0 = x_0 - x_0x_1 - c_0x_0w$ , $\dot{x}_1 = -x_1 + x_0x_1 - c_1x_1w$ , $\dot{x}_2 = (x_0 - 1)^2 + (x_1 - 1)^2$ , with $w(t) \in \{0, 1\}$                                             |
| <i>Initial conditions</i> | $x_0(0) = 0.5$ , $x_1(0) = 0.7$ , $x_2(0) = 0$                                                                                                                                                |
| <i>Path constraints</i>   | None written by the user; the convexification adds $\omega_0 + \omega_1 = 1$                                                                                                                  |
| <i>Bounds</i>             | $0 \leq x_0, x_1 \leq 2$ , $0 \leq x_2 \leq 100$ , $t_0 = 0$ , $t_f = 12$ both fixed. Bounds on a declared integer control are ignored: its admissible set governs                            |
| <i>Constants</i>          | $c_0 = 0.4$ , $c_1 = 0.2$                                                                                                                                                                     |
| <i>Discretisation</i>     | Trapezoidal, meshes of 40, 80, 160, 320 and 640 nodes solved in turn, automatic differentiation, automatic scaling, NLP tolerance $10^{-7}$                                                   |
| <i>Mesh refinement</i>    | None (fixed mesh); the sweep over meshes is driven by the example itself                                                                                                                      |
| <i>Known solution</i>     | No optimal integer control exists. The relaxed optimum $\Phi^* = 1.34408$ quoted by mintOC [17] is the infimum of the binary problem, approached but not attained                             |
| <i>Computed</i>           | At 640 nodes the relaxed optimum is 1.344093; the rounded binary schedule gives 1.344214, that is $1.339 \times 10^{-4}$ above $\Phi^*$ , with 77 switches                                    |
| <i>Source</i>             | <code>examples/lotka/lotka.cxx</code>                                                                                                                                                         |

## Results

The summary below is that of the last and finest solve of the sweep, the relaxed problem on 640 nodes.

#### PSOPT results summary

=====

Problem: Lotka-Volterra fishing (integer-control declaration API)

CPU time (seconds): 3.892406e+00

NLP solver used: IPOPT

Optimal (unscaled) cost function value: 1.344093e+00

Phase 1 endpoint cost function value: 1.344093e+00

Phase 1 integrated part of the cost: 0.000000e+00

Phase 1 initial time: 0.000000e+00

Phase 1 final time: 1.200000e+01

Phase 1 maximum relative local error: 2.417321e-04

NLP solver reports: The problem has been solved!

Table 12 collects the sweep. The relaxed objective in the second column is the optimum of the relaxation, a lower bound up to discretisation error on what any binary schedule can achieve; the third column is the objective of the rounded schedule, obtained by simulating it with a fixed-step Runge-Kutta integrator from the given initial state, and is therefore an achievable upper bound. The fourth column is the third minus  $\Phi^*$ .

Table 12: Lotka-Volterra fishing: the mesh sweep

| nodes | relaxed $\Phi$ | integer $\Phi$ | gap to $\Phi^*$ | switches |
|-------|----------------|----------------|-----------------|----------|
| 40    | 1.348102       | 1.351850       | 7.770e-03       | 4        |
| 80    | 1.344992       | 1.431325       | 8.724e-02       | 10       |
| 160   | 1.344232       | 1.354100       | 1.002e-02       | 20       |
| 320   | 1.344125       | 1.344667       | 5.867e-04       | 38       |
| 640   | 1.344093       | 1.344214       | 1.339e-04       | 77       |

## 12 An analytic test problem: integrated-residual transcription against collocation

This is a small scalar problem with a closed-form solution, introduced by Neuenhofen and Kerrigan [10] to compare direct transcription methods. One state, one control, a fixed horizon  $[0, \pi/2]$ , and an integrand  $y^2 + u \cos t$  that is quadratic in the state and linear in the control. The control bound  $-1.5 \leq u \leq 1$  never becomes active: the optimal control runs from  $u^*(0) = -1$  to  $u^*(\pi/2) = 0.125$ . The unconstrained minimum principle therefore holds throughout and the solution can be written down:  $\lambda^* = -\cos t$ ,  $y^*(t) = -\sin t/(2 - \cos t)$ ,  $u^*(t) = 1/2 - 1.5/(\cos t - 2)^2$ , with  $J^* = -0.256996962561$ . Every number a solver produces on this problem can therefore be compared against

the truth rather than against another solver, which is why it is used here to exercise *PSOPT*'s integrated-residual transcription.

Collocation forces the dynamic residual to zero at a finite set of points and says nothing whatever about the intervals between them. The discretised problem is consequently a *relaxation* of the continuous one, not a restriction of it, and its optimal value may lie below the true optimum. That is exactly what happens here: on 41 nodes, Hermite-Simpson returns  $J = -0.257079348996$ , which is  $8.24 \times 10^{-5}$  below  $J^*$ , a value no admissible trajectory can attain. The control that produces it is wrong by 1.68 in the maximum norm over the nodes, which is more than the entire range of  $u^*$ ; this is the familiar node-to-node ringing.

Neither of the two quantities one would naturally consult gives any hint of it. The objective is below the optimum, which looks like success. And *PSOPT*'s relative local error estimate reads  $3.35 \times 10^{-5}$ , which looks like an accurate solution. The estimate is not malfunctioning: it measures how nearly the transcription's own state and control polynomials satisfy the differential equation, and that pair, ringing control and all, satisfies it perfectly well. What has gone wrong is not feasibility. It is that Simpson's rule cannot integrate  $u(t) \cos t$  when  $u$  oscillates from one node to the next, so the cost the nonlinear program minimises is not quite the cost the user wrote; the same trajectory's polynomials, integrated accurately, give a cost  $1.9 \times 10^{-5}$  above  $J^*$ , on the admissible side. A residual that is small and a quadrature that is wrong look identical from the outside, which is the reason this example is here. The solve is also badly conditioned near its ringing solution and only certifies optimality at an NLP tolerance of  $10^{-5}$ .

Setting `transcription_method` to "integrated-residual" changes what is being constrained. *PSOPT* now evaluates the residual by Gauss-Legendre quadrature at `ir_residual_nodes` points in each interval, placed deliberately away from the mesh nodes, so that inter-node error is visible to the nonlinear program. The runs here use the well-conditioned residual-box form, `ir_objective` set to "cost" with a bound `ir_residual_bound` =  $\delta$ , which minimises the cost subject to  $|r| \leq \delta$  at every quadrature point, with  $\delta = h^p$  ( $h$  the node spacing,  $p$  the order of the local representation). This is still a relaxation, but a declared one, and the accounting is remarkably clean: over the flexible-order sequence the shortfall below  $J^*$  is very nearly proportional to  $\delta$ , the ratio  $|J - J^*|/\delta$  taking the values 40.6, 40.7, 40.6 and 44.9 as  $\delta$  falls through four decades. The declared bound thus buys an estimate of how far below the truth the reported cost can be, which collocation does not provide. With the Nie-Kerrigan representation [12, 11] at `ir_local_order` = 5, on the same 41 nodes as the collocation run,  $J = -0.257001152538$ : two orders of magnitude closer to  $J^*$ , a control error of  $1.54 \times 10^{-3}$ , and a relative local error of  $1.17 \times 10^{-7}$ .

Two things should be said against this. First, the default cubic-Hermite residual box on 41 nodes returns  $J = -0.304114780310$ , further from  $J^*$  than the collocation number is. That is the relaxation behaving as advertised rather than failing:  $\delta = h^2 = 1.5 \times 10^{-3}$  is a loose allowance, and since the cost depends on the control only through  $u \cos t$ , the optimiser spends the allowance where the weight  $\cos t$  is smallest, near  $t = \pi/2$ , where a large control excursion is nearly free. Second,  $\delta$  cannot be driven below what the



NLP can resolve. The  $p$ -sequence stops at order 5 because the next step would ask for  $\delta = h^6 \approx 10^{-9}$  against an NLP tolerance of  $10^{-7}$ .

Table 13: Neuenhofen-Kerrigan analytic problem: summary

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | Integrated-residual transcription: the residual box ( <code>ir_objective = "cost"</code> with <code>ir_residual_bound</code> ), the Nie-Kerrigan flexible-order representation, and the fact that collocation can report a cost below the true optimum                                                                                                                                                                                            |
| <i>Reference</i>           | Neuenhofen and Kerrigan [10]; the integrated-residual methods of Nie and Kerrigan [12, 11]                                                                                                                                                                                                                                                                                                                                                        |
| <i>Dimensions</i>          | 1 phase; 1 state, 1 control, 1 event, no path constraints, no static parameters                                                                                                                                                                                                                                                                                                                                                                   |
| <i>Objective</i>           | $J = \int_0^{\pi/2} [y(t)^2 + u(t) \cos t] dt$                                                                                                                                                                                                                                                                                                                                                                                                    |
| <i>Dynamics</i>            | $\dot{y} = \frac{1}{2}y^2 + u$                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <i>Initial conditions</i>  | $y(0) = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <i>Terminal conditions</i> | None; $y(t_f)$ is free                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <i>Bounds</i>              | $-10 \leq y \leq 10$ , $-1.5 \leq u \leq 1$ (inactive at the solution), $t_0 = 0$ and $t_f = \pi/2$ both fixed                                                                                                                                                                                                                                                                                                                                    |
| <i>Discretisation</i>      | Hermite-Simpson, automatic differentiation, automatic scaling, IPOPT; 41 nodes for the three-way comparison and $8p + 1$ nodes for the $p$ -sequence; <code>ir_residual_scaling = "state"</code> , <code>ir_residual_bound = h^p</code> , <code>ir_residual_nodes = 4</code> for the cubic-Hermite runs and $2p$ for the flexible-order runs; NLP tolerance $10^{-5}$ for the two 41-node comparison runs and $10^{-7}$ for the Nie-Kerrigan runs |
| <i>Mesh refinement</i>     | None (fixed mesh); the mesh and the local order are varied by hand between solves                                                                                                                                                                                                                                                                                                                                                                 |
| <i>Known solution</i>      | $y^*(t) = -\sin t/(2 - \cos t)$ , $u^*(t) = 1/2 - 1.5/(\cos t - 2)^2$ , $\lambda^*(t) = -\cos t$ , $J^* = -0.256996962561$ , stated by the source and verified there by quadrature to about $10^{-14}$                                                                                                                                                                                                                                            |
| <i>Computed</i>            | On 41 nodes: collocation $J = -0.257079348996$ ( $8.24 \times 10^{-5}$ below $J^*$ , control error 1.68); cubic-Hermite residual box $J = -0.304114780310$ ; Nie-Kerrigan order 5 $J = -0.257001152538$ ( $4.19 \times 10^{-6}$ from $J^*$ , control error $1.54 \times 10^{-3}$ , relative local error $1.17 \times 10^{-7}$ )                                                                                                                   |
| <i>Source</i>              | <code>examples/x1/x1_example.cxx</code>                                                                                                                                                                                                                                                                                                                                                                                                           |

## Results

The example performs six solves of the same problem and prints its own comparison against the analytic solution, reproduced below; it runs with `print_level` set to zero and writes no options or summary file of its own, so the table is the whole of the result. The last two columns are the ones to compare:  $\max |u - u^*|$  is the largest control error

at a node, and `relLocErr` is *PSOPT*'s estimate of the relative local error.

```
X1 -- Neuenhofen-Kerrigan analytic example (J* = -0.256996962561)
=====
method          nodes          J      |J-J*|  max|u-u*|  relLocErr
-----
collocation (Hermite-Simpson)  41   -0.257079348996  8.24e-05  1.68e+00  3.35e-05
integrated-residual (cubic-H)  41   -0.304114780310  4.71e-02  1.52e+00  8.74e-04
Nie-Kerrigan order 5          41   -0.257001152538  4.19e-06  1.54e-03  1.17e-07
-----
Nie-Kerrigan flexible-order p-convergence (8 elements, order-aware box):
order 2          17   -0.647590089554  3.91e-01  4.28e-01  1.63e-02
order 3          25   -0.268360081463  1.14e-02  2.46e-02  4.93e-04
order 4          33   -0.257232674838  2.36e-04  3.09e-03  7.56e-06
order 5          41   -0.257001152538  4.19e-06  1.54e-03  1.17e-07
=====
Collocation reports J below J* (infeasible between nodes) and a ringing control;
integrated-residual is feasibility-honest; flexible-order p-refinement converges
to the analytic optimum.
```

### 13 Chance-constrained double integrator: a probabilistic path constraint by constraint tightening

A chance constraint asks that a path constraint hold with at least a stated probability,  $P\{v(t) \leq V_{\max}\} \geq 1 - \epsilon$ , rather than with certainty. The oldest and cheapest way to impose one is to convert it into a deterministic constraint on the nominal trajectory: if the estimate of  $v(t)$  is Gaussian with standard deviation  $\sigma$ , the requirement is met exactly when the nominal value satisfies  $v(t) + k\sigma \leq V_{\max}$  with  $k = \Phi^{-1}(1 - \epsilon)$ , so the original limit is tightened by the back-off  $k\sigma$ . That is all this example does, and it is worth being clear about what it does not do. It does not propagate uncertainty:  $\sigma$  is a constant put in by hand purely to illustrate the mechanism, the Gaussian assumption is asserted rather than derived, and the  $1 - \epsilon$  guarantee obtained is pointwise in time rather than over the whole trajectory. The comments in the source say what a full treatment would require: carry  $\sigma(t)$  as extra states governed by a Lyapunov differential equation driven by the linearised dynamics and the process noise, and evaluate the back-off from those states. *PSOPT* is perfectly able to solve that larger problem, since the extra covariance states are ordinary states and the tightened constraint is an ordinary path constraint, but it is not the problem solved here. The problem is a constructed illustration and not taken from the literature.

The nominal problem is the textbook minimum-energy move of a double integrator:  $\dot{x} = v$ ,  $\dot{v} = u$ , from  $(x, v) = (0, 0)$  to  $(1, 0)$  in one unit of time, minimising  $\int_0^1 u^2 dt$ . Its unconstrained solution is  $u = 6 - 12t$ ,  $v = 6t(1 - t)$ ,  $x = 3t^2 - 2t^3$ , with cost 12 and a velocity that peaks at 1.5 half way through. The numbers in the example are chosen so that this peak is inadmissible: with  $V_{\max} = 1.30$ ,  $\sigma = 0.03$  and  $k = 1.6449$  (that is,  $\epsilon = 5$  per cent) the tightened limit is  $1.30 - 1.6449 \times 0.03 = 1.250653$ , well below 1.5, so the constraint binds over an interval in the middle of the move. The computed cost is 13.87380, some 16 per cent more energy than the unconstrained move, which is the price of the probabilistic margin. A little over half of that price is charged by the nominal

limit alone: enforcing  $v \leq 1.30$  without any back-off would cost 13.0193, so the  $k\sigma$  term accounts for about 45 per cent of the increase.

The constrained problem is simple enough to solve by hand, which makes it a useful check. The solution is a boundary arc on which  $v$  sits exactly on the tightened limit  $V_t$  and  $u = 0$ , joined to entry and exit arcs on which  $u$  is linear in time; matching the distance travelled gives an entry time  $t_1 = \frac{3}{2}(1 - 1/V_t) = 0.30063$ , a peak magnitude  $2V_t/t_1 = 8.32$  for the control, and a cost  $16V_t^3/(9(V_t - 1)) = 13.87443$ . The computed value is below this by  $4.5 \times 10^{-5}$  in relative terms, that is, very slightly better than optimal, which is what one expects when an active path constraint is imposed only at the collocation nodes and may be exceeded a little between them.

The maximum relative local error,  $8.9 \times 10^{-10}$  on a fixed grid of 40 Legendre nodes, is worth a moment's thought, because it is smaller than the error in the answer. The estimate measures how nearly the transcription's own state and control polynomials satisfy the differential equations, and on this problem that pair is very nearly consistent: it is a smooth solution of the dynamics, and the dynamics here are linear. What it is not is the true optimal control, which has corners where the boundary arc is entered and left and which no single global polynomial can represent; the cost is  $4.5 \times 10^{-5}$  below the analytic optimum for exactly that reason. The two quantities answer different questions, and the discretisation error answers only the first of them. Local collocation with automatic mesh refinement, as used in Sections 8 and 15, would place nodes near the junctions and bring the cost closer, which is the quantity that has room to improve.

## Results

```
PSOPT results summary
=====

Problem:  Chance-constrained double integrator
CPU time (seconds): 4.731640e-01
NLP solver used:  IPOPT

Optimal (unscaled) cost function value: 1.387380e+01
Phase 1 endpoint cost function value: 0.000000e+00
Phase 1 integrated part of the cost: 1.387380e+01
Phase 1 initial time: 0.000000e+00
Phase 1 final time: 1.000000e+00
Phase 1 maximum relative local error: 8.851978e-10
NLP solver reports:  The problem has been solved!
```

The run is short: 51 IPOPT iterations and 0.473 seconds in total, of which 0.390 seconds is the NLP solve. The residual difference from the analytic optimum is the discretisation of the junctions and not a failure of convergence; IPOPT reports the problem solved,

Table 14: Chance-constrained double integrator: summary

|                            |                                                                                                                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | A probabilistic path constraint imposed by deterministic tightening: the nominal bound is reduced by the back-off $k\sigma$ with $k = \Phi^{-1}(1-\epsilon)$                                                                                                                                   |
| <i>Reference</i>           | None. The problem is a constructed illustration                                                                                                                                                                                                                                                |
| <i>Dimensions</i>          | 1 phase; 2 states, 1 control, 4 events, 1 path constraint, no static parameters                                                                                                                                                                                                                |
| <i>Objective</i>           | $J = \int_0^1 u^2 dt$                                                                                                                                                                                                                                                                          |
| <i>Dynamics</i>            | $\dot{x} = v, \quad \dot{v} = u$                                                                                                                                                                                                                                                               |
| <i>Initial conditions</i>  | $x(0) = 0, \quad v(0) = 0, \quad t_0 = 0$                                                                                                                                                                                                                                                      |
| <i>Terminal conditions</i> | $x(t_f) = 1, \quad v(t_f) = 0, \quad t_f = 1$                                                                                                                                                                                                                                                  |
| <i>Path constraints</i>    | $v + k\sigma - V_{\max} \leq 0$ , the deterministic equivalent of $P\{v \leq V_{\max}\} \geq 1 - \epsilon$ for a Gaussian velocity estimate of standard deviation $\sigma$                                                                                                                     |
| <i>Bounds</i>              | $-5 \leq x, v \leq 5, \quad -20 \leq u \leq 20, \quad t_0 = 0, t_f = 1$ both fixed                                                                                                                                                                                                             |
| <i>Constants</i>           | $V_{\max} = 1.30, \quad \sigma = 0.03$ (constant and illustrative; not propagated), $k = \Phi^{-1}(0.95) = 1.6449$ , giving $\epsilon = 5$ per cent and a tightened limit $V_t = 1.250653$                                                                                                     |
| <i>Discretisation</i>      | Legendre pseudospectral, 40 nodes, automatic differentiation, automatic scaling, NLP tolerance $10^{-6}$                                                                                                                                                                                       |
| <i>Mesh refinement</i>     | None (fixed mesh)                                                                                                                                                                                                                                                                              |
| <i>Known solution</i>      | No published value. Analytically, the unconstrained move costs 12 with $v_{\max} = 1.5$ ; with the boundary arc active the cost is $16V_t^3/(9(V_t - 1)) = 13.87443$ , the boundary arc runs from $t_1 = \frac{3}{2}(1 - 1/V_t) = 0.30063$ to $1 - t_1$ , and $ u $ peaks at $2V_t/t_1 = 8.32$ |
| <i>Computed</i>            | $J = 1.387380 \times 10^1$ , below the analytic value by $4.5 \times 10^{-5}$ in relative terms; maximum relative local error $8.851978 \times 10^{-10}$                                                                                                                                       |
| <i>Source</i>              | <code>examples/chance_constraint/chance_constraint.cxx</code>                                                                                                                                                                                                                                  |

and the objective is unchanged in all eight printed digits,  $1.3873797 \times 10^1$ , over the final twenty iterations.

## 14 Manutec r3 robot: a multibody model and *hp*-adaptive mesh refinement

Most of the plants in this document are written out in a few lines of algebra. This one is not. The dynamics used here are the DLR model 2 of the Manutec r3 industrial robot, due to Otter and Turk [13] and documented as a benchmark by Franke and Otter [6]; von Stryk and Schlemmer [20] used it as an optimal control test case. It covers the first three links, with the equation of motion

$$\mathbf{M}(\mathbf{q}(t)) \ddot{\mathbf{q}}(t) = \mathbf{V}(\mathbf{q}(t), \dot{\mathbf{q}}(t)) + \mathbf{G}(\mathbf{q}(t)) + \mathbf{D} \mathbf{u}(t)$$

where  $\mathbf{q} = [q_1 \ q_2 \ q_3]^T$  are the relative joint angles,  $\mathbf{u}$  are the three motor voltages,  $\mathbf{M}$  is the symmetric inertia matrix,  $\mathbf{V}$  collects the Coriolis and centrifugal torques,  $\mathbf{G}$  the gravitational torques, and  $\mathbf{D} = \text{diag}(-126, 252, 72)$  N m/V converts voltages into generalised forces. None of these terms is a textbook idealisation: they carry the measured link masses, lengths, centre of mass offsets and inertia tensors of the real machine, and they are evaluated by a routine translated from the DLR simulation code and shipped in full with the example<sup>1</sup>, which the user function `dae` does no more than call. A mechanical model of realistic complexity can therefore be handed to *PSOPT* as a black box, with automatic differentiation taking derivatives through it and no Jacobian written by hand.

The manoeuvre posed here is a point to point transfer in fixed time: move from rest at  $\mathbf{q}(0) = (0, -1.5, 0)$  to rest at  $\mathbf{q}(t_f) = (1, -1.95, 1)$  radians in  $t_f = 0.53$  s while minimising the integral of  $\mathbf{u}^T \mathbf{u}$ , subject to the machine's joint angle, joint rate and voltage limits. The horizon is tight enough that the first motor cannot be driven gently: the computed  $u_1$  sits on its lower limit of  $-7.5$  V for roughly the first 0.035 s, sweeps through zero and returns to  $+7.5$  V at the final time, while  $u_2$  and  $u_3$  stay comfortably inside their bounds (Figure 19). The joint rate limits are approached but never reached,  $\dot{q}_1$  peaking just under its 3 rad/s bound.

That saturated arc is what makes the example the natural place to look at *hp*-adaptive refinement. Setting `algorithm.mesh_refinement = "automatic"` with a global Legendre discretisation hands the mesh to *PSOPT*'s *hp* driver, which measures the decay of the Legendre coefficients on each interval and raises the polynomial degree where the solution looks smooth, splitting an interval only where raising the degree has stopped paying or the degree cap (here  $N_{\max} = 12$ ) has been reached; growth is limited to forty per cent of the node count per iteration.

As shipped, however, the driver has nothing to do. The 20 nodes given in the source already meet the requested ODE tolerance of  $10^{-5}$ , at  $1.34 \times 10^{-6}$ , so the refinement

---

<sup>1</sup>Dr Martin Otter of DLR, Germany, has kindly authorised the author to publish a translated form of subroutine R3M2SI as part of the *PSOPT* distribution.

stops after one pass and the remaining entries of the source’s node list are never used. To see the mechanism work, set `algorithm.ode_tolerance` to  $10^{-9}$ : the driver then chooses 21, 25, 35 and finally 46 nodes, and the error estimate falls monotonically through  $1.34 \times 10^{-6}$ ,  $5.11 \times 10^{-7}$ ,  $6.88 \times 10^{-8}$ ,  $2.63 \times 10^{-8}$  and  $3.51 \times 10^{-9}$  — three orders of magnitude for rather more than twice the nodes, which is what  $p$ -refinement on a smooth solution should look like.

The objective is the reassuring part of that experiment. It reads  $2.040442 \times 10^1$  on the first mesh and  $2.040446 \times 10^1$  on the last, two parts in a million apart, so the first mesh was converged and the estimate was right to stop. Contrast the minimum time to climb problem of §15, where a low order scheme is held fixed and accuracy is bought purely by adding nodes where the error is largest.

## Results

```
PSOPT results summary
=====

Problem:  Manutec r3 robot problem
CPU time (seconds): 4.475920e-01
NLP solver used:  IPOPT

Optimal (unscaled) cost function value:  2.040442e+01
Phase 1 endpoint cost function value: 0.000000e+00
Phase 1 integrated part of the cost:  2.040442e+01
Phase 1 initial time: 0.000000e+00
Phase 1 final time: 5.300000e-01
Phase 1 maximum relative local error: 1.336018e-06
NLP solver reports:  The problem has been solved!
```

The single-row refinement table is in Table 16. The solve is cheap: 860 evaluations of the right-hand side and 0.45 s in total, of which 0.04 s was spent by *PSOPT* outside the NLP solver. The Hessian evaluation count is zero because IPOPT was run in limited-memory mode.

## 15 Minimum time to climb: a model driven by look-up tables

A supersonic aircraft starts at sea level at 424.26 ft/s and must reach 65 600 ft at 968.148 ft/s in as little time as possible, with the flight path angle zero at both ends and the angle of attack as the only control. The cost is the final time itself and nothing else, so the entire problem is carried by the dynamics; the brachistochrone of §2 is the analytic minimum-time problem of this document and this is the practical one. It is

Table 15: Manutec r3 robot: summary

|                            |                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | A realistic multibody model used as a black box within <code>dae</code> , and automatic <i>hp</i> -adaptive mesh refinement of a global pseudospectral discretisation                                                                                                                                                                                             |
| <i>Reference</i>           | DLR model 2 of Otter and Turk [13]; benchmark description by Franke and Otter [6]; optimal control of the same robot by von Stryk and Schlemmer [20]                                                                                                                                                                                                              |
| <i>Dimensions</i>          | 1 phase; 6 states, 3 controls, 12 events, no path constraints, no static parameters                                                                                                                                                                                                                                                                               |
| <i>Objective</i>           | $J = \int_0^{t_f} \mathbf{u}(t)^T \mathbf{u}(t) dt$ with $t_f$ fixed (OBJ_OPTION 3; the source also offers minimum time and regularised minimum time variants)                                                                                                                                                                                                    |
| <i>Dynamics</i>            | $\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} = \mathbf{V}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{G}(\mathbf{q}) + \mathbf{D}\mathbf{u}$ , written as six first order equations in $\mathbf{x} = [\mathbf{q}^T \dot{\mathbf{q}}^T]^T$ . The inertia, Coriolis and gravity terms are evaluated by the translated DLR routine <code>r3m2si</code> in the example source |
| <i>Initial conditions</i>  | $\mathbf{q}(0) = (0, -1.5, 0) \text{ rad}, \quad \dot{\mathbf{q}}(0) = \mathbf{0}$                                                                                                                                                                                                                                                                                |
| <i>Terminal conditions</i> | $\mathbf{q}(t_f) = (1, -1.95, 1) \text{ rad}, \quad \dot{\mathbf{q}}(t_f) = \mathbf{0}$                                                                                                                                                                                                                                                                           |
| <i>Bounds</i>              | $ q_i  \leq (2.97, 2.01, 2.86) \text{ rad}, \quad  \dot{q}_i  \leq (3.0, 1.5, 5.2) \text{ rad/s}, \quad  u_i  \leq 7.5 \text{ V}, \quad t_0 = 0, \quad t_f = 0.53 \text{ s}$                                                                                                                                                                                      |
| <i>Constants</i>           | $g = 9.81 \text{ m/s}^2$ , zero payload, and the DLR link and motor data (masses, lengths, centres of mass, inertia tensors, gear ratios, voltage-force constants) listed at the head of <code>main</code> in the source                                                                                                                                          |
| <i>Discretisation</i>      | Legendre pseudospectral with the standard differentiation matrix, 20 nodes initially, automatic differentiation, automatic scaling, state-based defect scaling, NLP tolerance $10^{-6}$                                                                                                                                                                           |
| <i>Mesh refinement</i>     | Automatic, <i>hp</i> ( <i>ph</i> ) variant for global pseudospectral methods; ODE tolerance $10^{-5}$ , at most 5 iterations, growth capped at 0.4, interval orders in [3, 12]; the tolerance is met on the initial 20 nodes, so no refinement takes place                                                                                                        |
| <i>Known solution</i>      | None: neither the example source nor this document quotes a published value of the minimum energy cost for these boundary conditions and this horizon                                                                                                                                                                                                             |
| <i>Computed</i>            | $J = 2.040442 \times 10^1$ , with a maximum relative local error of $1.336018 \times 10^{-6}$ against a requested $10^{-5}$ , in 0.45 s of CPU time                                                                                                                                                                                                               |
| <i>Source</i>              | <code>examples/manutec/manutec.cxx</code>                                                                                                                                                                                                                                                                                                                         |

Table 16: Mesh refinement statistics: Manutec r3 robot problem

| Iter             | DM     | M  | NV  | NC  | OE | CE | JE | HE | RHS | $\epsilon_{\max}$ | CPU <sub>a</sub> |
|------------------|--------|----|-----|-----|----|----|----|----|-----|-------------------|------------------|
| 1                | LGL-ST | 20 | 182 | 133 | 43 | 43 | 35 | 0  | 860 | 1.336e-06         | 4.071e-01        |
| CPU <sub>b</sub> | -      | -  | -   | -   | -  | -  | -  | -  | -   | -                 | 4.045e-02        |
| -                | -      | -  | -   | -   | 43 | 43 | 35 | 0  | 860 | -                 | 4.476e-01        |

*Key:* Iter=iteration number, DM= discretization method, M=number of nodes, NV=number of variables, NC=number of constraints, OE=objective evaluations, CE = constraint evaluations, JE = Jacobian evaluations, HE = Hessian evaluations, RHS = ODE right hand side evaluations,  $\epsilon_{\max}$  = maximum relative ODE error, CPU<sub>a</sub> = CPU time in seconds spent by NLP algorithm, CPU<sub>b</sub> = additional CPU time in seconds spent by PSOPT

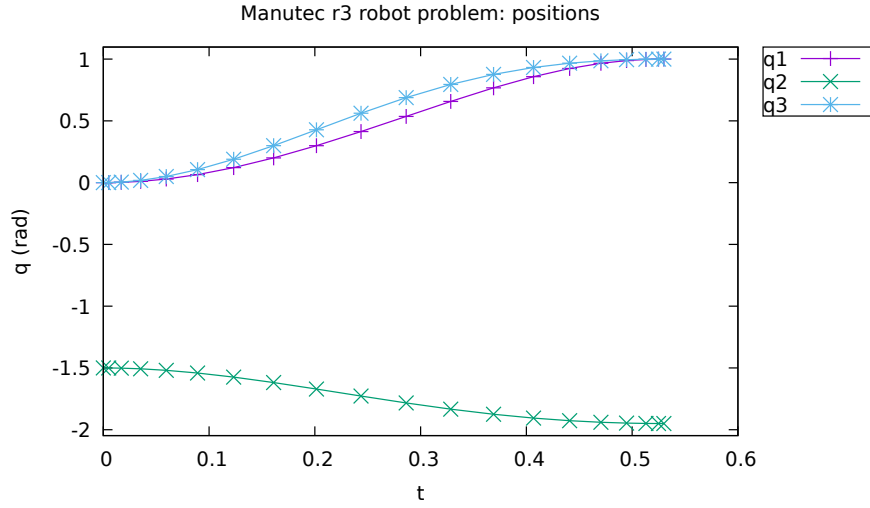


Figure 17: Joint angles  $q_1$ ,  $q_2$  and  $q_3$  for the Manutec r3 minimum energy problem



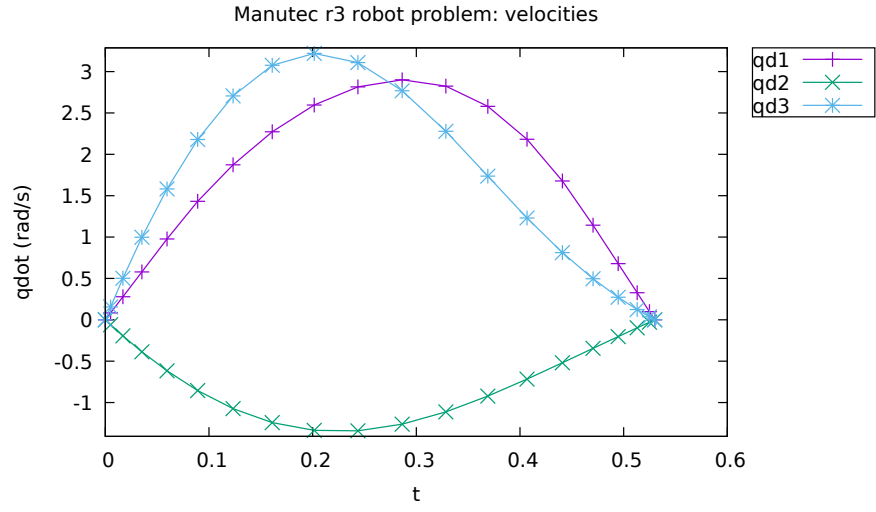


Figure 18: Joint rates  $\dot{q}_1$ ,  $\dot{q}_2$  and  $\dot{q}_3$  for the Manutec r3 minimum energy problem

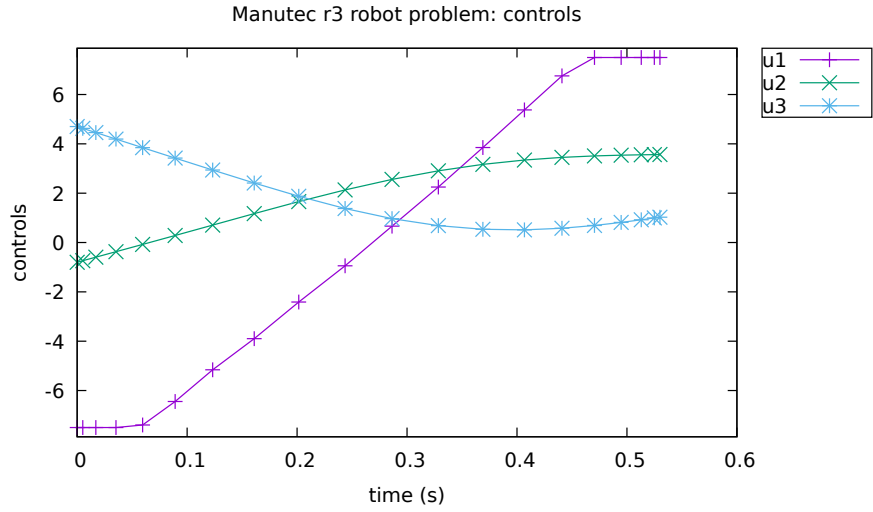


Figure 19: Motor voltages  $u_1$ ,  $u_2$  and  $u_3$  for the Manutec r3 minimum energy problem. The first control is saturated at  $-7.5$  V over the initial arc and returns to  $+7.5$  V at the final time

a long-standing benchmark for trajectory optimisation software: it descends from the energy-state analysis of supersonic climb by Bryson, Desai and Hoffman [3], is treated at length by Betts [2], and is taken here in the form given in the GPOPS user’s manual [14].

What distinguishes it from the other examples is that the model is not written in closed form. The lift-curve slope  $c_{L\alpha}$ , the zero-lift drag coefficient  $c_{D0}$  and the induced drag factor  $\eta$  are each obtained by one-dimensional spline interpolation of nine tabulated values against Mach number, and the thrust  $T(M, h)$  by two-dimensional spline interpolation of a  $10 \times 10$  table against Mach number and altitude. Air density and the speed of sound come from the 1976 standard atmosphere, through the library’s `include/us76.h`, whose lower branch is the closed-form solution of the standard’s layer equations written without any comparison that a derivative tape could freeze. The right-hand side is therefore only as smooth as the spline interpolation of the aerodynamic tables makes it, and that shows in the results: the maximum relative local error starts at  $3.6 \times 10^{-2}$  and reaches only  $2.7 \times 10^{-4}$  over three mesh refinement iterations, where a problem with an analytic right-hand side of the same size would have reached  $10^{-9}$  (compare §2).

A model of this kind raises a difficulty that is easy to miss and is worth stating plainly, because it applies to any look-up table used under automatic differentiation. Selecting an atmospheric layer, or a spline interval, means comparing a quantity that depends on the states against a table of breakpoints. An ordinary C++ comparison on a taped quantity is resolved once, when the derivative tape is recorded, and is then frozen: the layer or the interval stays at whatever it was at the point of taping, and the model silently becomes wrong as soon as the trajectory moves out of it.

There are two ways out, and this example has used both. The first is to record the comparison on the tape, with the conditional operations *PSOPT* provides: every layer’s formula is evaluated and `psopt_cond_lt` selects among them, so the choice is re-made at every evaluation. That works, and it is what *PSOPT*’s interpolation routines do for their spline intervals, but it has a sharp edge — an unselected layer’s extrapolated temperature goes negative (the tropospheric lapse rate reaches 0 K near 44 km) and a negative base in the pressure power law puts a NaN on the tape even in the arm that is discarded, so the local temperature has to be floored.

The second is to have no comparison to record. The atmosphere now comes from the library’s `include/us76.h`, which writes the temperature as the sum of the lapse rates against smooth positive parts and the pressure as the closed-form solution of hydrostatic equilibrium through those same layers: one expression, no branches, and a kilometre clear of a layer boundary it reproduces the layer equations to two parts in  $10^5$ . Where the selection can be removed rather than taped, removing it is the better answer, and it leaves one implementation of the standard atmosphere in the distribution instead of three. The altitude is still passed in as the taped variable rather than through `.value()`, which would leave the density and the speed of sound with no sensitivity to altitude at all.

The computed minimum time is 321.01 s, and the trajectory has the shape expected of a supersonic climb: about 30 s accelerating in level flight, a climb to roughly 31 000 ft,

a shallow dive back to about 26 000 ft in which the aircraft accelerates through the transonic drag rise, and then a zoom climb during which the speed falls from its peak near 1700 ft/s to the required 968.148 ft/s. The angle of attack stays well inside its  $\pm 20^\circ$  bounds, between  $5.8^\circ$  and  $-6.6^\circ$ , so no bound is active.

## Results

```
PSOPT results summary
=====

Problem: Minimum time to climb for a supersonic aircraft
CPU time (seconds): 8.316002e+00
NLP solver used: IPOPT

Optimal (unscaled) cost function value: 3.210142e+02
Phase 1 endpoint cost function value: 3.210142e+02
Phase 1 integrated part of the cost: 0.000000e+00
Phase 1 initial time: 0.000000e+00
Phase 1 final time: 3.210142e+02
Phase 1 maximum relative local error: 2.709068e-04
NLP solver reports: The problem has been solved!
```

The mesh refinement history is in Table 18. It is the document’s example of local, or  $h$ -type, refinement: the discretisation is a fixed low-order collocation scheme and accuracy is bought by adding nodes where the estimated error is largest and, once, by raising the order of the scheme from trapezoidal to Hermite-Simpson after the second iteration. This is the opposite of the  $hp$ -adaptive strategy used for the Manutec robot in §14, where the polynomial degree within each interval is the variable of first resort and the number of intervals changes only when raising the degree stops paying.

The refinement stopped on its tolerance, at 58 nodes and a maximum relative local error of  $2.7 \times 10^{-4}$  against a requested  $10^{-3}$ , having taken 30 765 evaluations of the right-hand side and 8.3 s. Two of the three iterations bought very little —  $3.6 \times 10^{-2}$  to  $1.5 \times 10^{-2}$  for twelve more nodes — and the factor of fifty at the third came from the change of scheme rather than from the mesh. The slow decrease under  $h$ -refinement alone is the interpolation showing through, as discussed above, and not a defect of the refinement algorithm.

## 16 Multiphase launch vehicle ascent: linkages, mass discontinuities and terminal orbital elements

This is the largest application in the document: the ascent of a launch vehicle from Cape Canaveral into a geosynchronous transfer orbit, taken from the GPOPS documentation

Table 17: Minimum time to climb: summary

|                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>                                                                                                                                                                                                                                                                                               | A minimum-time problem whose model is driven by tabulated data: 1-D and 2-D spline interpolation, tape-safe interval selection under automatic differentiation, and local ( $h$ ) mesh refinement with a change of collocation method                                                           |
| <i>Reference</i>                                                                                                                                                                                                                                                                                                  | GPOPS user's manual [14]; also in Betts [2]                                                                                                                                                                                                                                                     |
| <i>Dimensions</i>                                                                                                                                                                                                                                                                                                 | 1 phase; 4 states, 1 control, 7 events, no path constraints, no static parameters                                                                                                                                                                                                               |
| <i>Objective</i>                                                                                                                                                                                                                                                                                                  | $J = t_f$                                                                                                                                                                                                                                                                                       |
| <i>Dynamics</i>                                                                                                                                                                                                                                                                                                   | $\dot{h} = v \sin \gamma, \quad \dot{w} = -T(M, h)/I_{sp},$ $\dot{v} = \frac{1}{m} [T(M, h) \cos \alpha - D] - \frac{\mu}{(R_e + h)^2} \sin \gamma,$ $\dot{\gamma} = \frac{1}{mv} [T(M, h) \sin \alpha + L] + \cos \gamma \left[ \frac{v}{R_e + h} - \frac{\mu}{v(R_e + h)^2} \right]$          |
| with $m = w/g_0$ ,<br>$M = v/c$ ,<br>$c = 20.0468\sqrt{\theta}$ m/s,<br>$D = \frac{1}{2}C_D S \rho v^2$ ,<br>$L = \frac{1}{2}C_L S \rho v^2$ ,<br>$C_L = c_{L\alpha}(M) \alpha$ ,<br>$C_D = c_{D0}(M) + \eta(M) c_{L\alpha}(M) \alpha^2$ ,<br>and $\rho(h)$ , $\theta(h)$ from<br>the 1976 standard<br>atmosphere |                                                                                                                                                                                                                                                                                                 |
| <i>Initial conditions</i>                                                                                                                                                                                                                                                                                         | $h(0) = 0$ ft, $v(0) = 424.26$ ft/s, $\gamma(0) = 0$ , $w(0) = 42\,000$ lb,<br>$t_0 = 0$                                                                                                                                                                                                        |
| <i>Terminal conditions</i>                                                                                                                                                                                                                                                                                        | $h(t_f) = 65\,600$ ft, $v(t_f) = 968.148$ ft/s, $\gamma(t_f) = 0$ ; $w(t_f)$ free                                                                                                                                                                                                               |
| <i>Bounds</i>                                                                                                                                                                                                                                                                                                     | $0 \leq h \leq 69\,000$ ft, $1 \leq v \leq 2000$ ft/s, $ \gamma  \leq 89^\circ$ , $0 \leq w \leq 45\,000$ lb, $ \alpha  \leq 20^\circ$ , $200 \leq t_f \leq 500$ s                                                                                                                              |
| <i>Constants</i>                                                                                                                                                                                                                                                                                                  | $S = 530$ ft <sup>2</sup> , $I_{sp} = 1600$ s, $g_0 = 32.174$ ft/s <sup>2</sup> , $R_e = 2.09029 \times 10^7$ ft, $\mu = 0.14076539 \times 10^{17}$ ft <sup>3</sup> /s <sup>2</sup> ; sea-level standard density 1.22521 kg/m <sup>3</sup> and temperature 288.15 K                             |
| <i>Data</i>                                                                                                                                                                                                                                                                                                       | Tabulated aerodynamic and propulsion data, listed in the source and not reproduced here: $c_{L\alpha}$ , $c_{D0}$ and $\eta$ at nine Mach numbers from 0 to 1.8, and thrust at ten Mach numbers by ten altitudes from 0 to 70 000 ft; and the eight-layer table of the 1976 standard atmosphere |
| <i>Discretisation</i>                                                                                                                                                                                                                                                                                             | Trapezoidal, 30 nodes initially, automatic differentiation, automatic scaling, Jacobian-based defect scaling, limited-memory Hessian, NLP tolerance $10^{-6}$                                                                                                                                   |
| <i>Mesh refinement</i>                                                                                                                                                                                                                                                                                            | Automatic, local ( $h$ ) variant; ODE tolerance $10^{-3}$ , at most 4 iterations, order switched to Hermite-Simpson after the second; 30 nodes grow to 58                                                                                                                                       |
| <i>Known solution</i>                                                                                                                                                                                                                                                                                             | None: neither the example source nor this document quotes a published value of the minimum time                                                                                                                                                                                                 |
| <i>Computed</i>                                                                                                                                                                                                                                                                                                   | $t_f = 3.210142 \times 10^2$ s, with a maximum relative local error of $2.709068 \times 10^{-4}$ and 8.3 s of CPU time                                                                                                                                                                          |
| <i>Source</i>                                                                                                                                                                                                                                                                                                     | <code>examples/climb/climb.cxx</code>                                                                                                                                                                                                                                                           |

Table 18: Mesh refinement statistics: Minimum time to climb for a supersonic aircraft

| Iter             | DM  | M  | NV  | NC  | OE  | CE  | JE  | HE | RHS   | $\epsilon_{\max}$ | CPU <sub>a</sub> |
|------------------|-----|----|-----|-----|-----|-----|-----|----|-------|-------------------|------------------|
| 1                | TRP | 30 | 152 | 128 | 42  | 42  | 41  | 0  | 2478  | 3.553e-02         | 9.683e-01        |
| 2                | TRP | 42 | 212 | 176 | 44  | 45  | 43  | 0  | 3735  | 1.471e-02         | 1.378e+00        |
| 3                | H-S | 58 | 349 | 240 | 106 | 107 | 71  | 0  | 24552 | 2.709e-04         | 5.753e+00        |
| CPU <sub>b</sub> | -   | -  | -   | -   | -   | -   | -   | -  | -     | -                 | 2.170e-01        |
| -                | -   | -  | -   | -   | 192 | 194 | 155 | 0  | 30765 | -                 | 8.316e+00        |

*Key:* Iter=iteration number, DM= discretization method, M=number of nodes, NV=number of variables, NC=number of constraints, OE=objective evaluations, CE = constraint evaluations, JE = Jacobian evaluations, HE = Hessian evaluations, RHS = ODE right hand side evaluations,  $\epsilon_{\max}$  = maximum relative ODE error, CPU<sub>a</sub> = CPU time in seconds spent by NLP algorithm, CPU<sub>b</sub> = additional CPU time in seconds spent by PSOPT

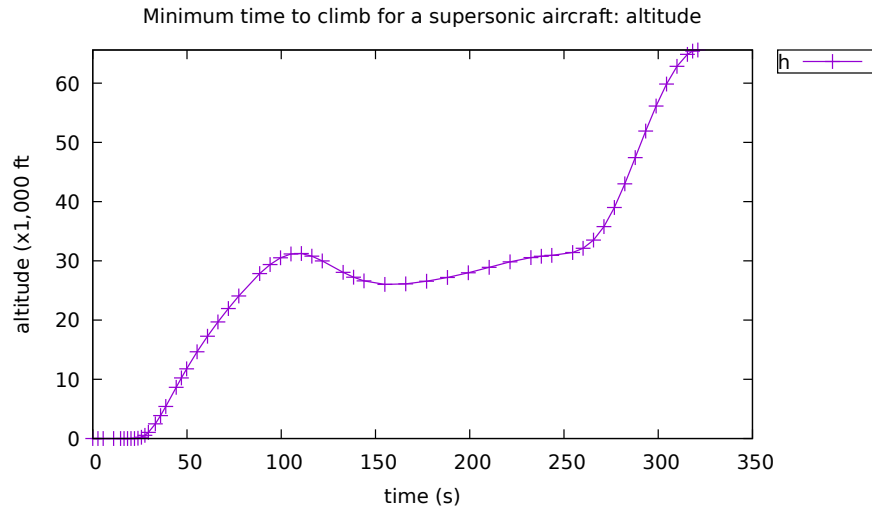


Figure 20: Altitude for the minimum time to climb problem

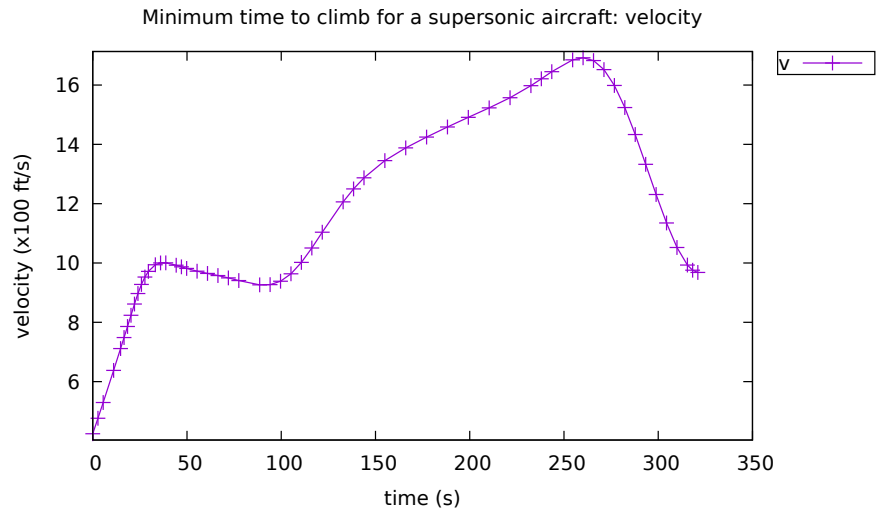


Figure 21: Velocity for the minimum time to climb problem

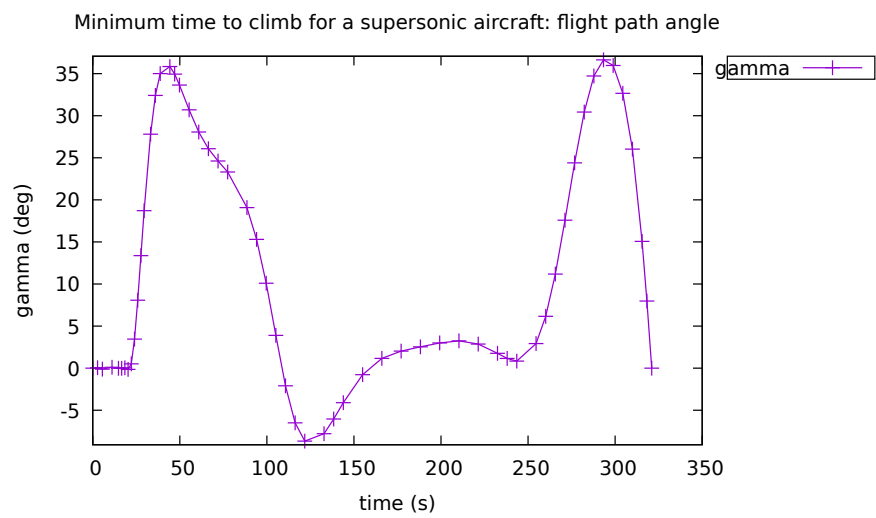


Figure 22: Flight path angle for the minimum time to climb problem

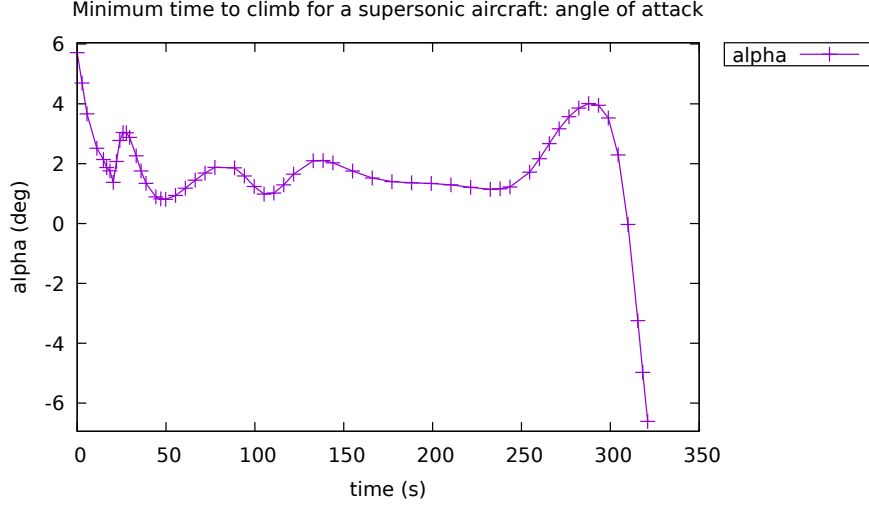


Figure 23: Angle of attack for the minimum time to climb problem

[14] and from Benson’s thesis [1]. The vehicle carries nine strap-on solid boosters, a liquid first stage, a second stage and a payload of 4164 kg, for a lift-off mass of 301 454 kg. Six boosters are lit at lift-off and burn for 75.2 s; the remaining three are lit as the first six are jettisoned and burn until 150.4 s; the first stage then continues alone until 261 s; the second stage flies the rest of the ascent. Those four intervals are the four phases. Each has seven states (Earth-centred inertial position and velocity, and mass) and three controls, the components of the thrust direction, constrained by the path condition  $\|\mathbf{u}\|^2 = 1$ . The first three phase durations are fixed by the burn times; the final time of phase 4 is free within [261, 961] s, and the objective is to maximise the mass delivered, posed as the minimisation of  $-m^{(4)}(t_f)$ .

The linkage machinery introduced in 3 does real work here. Each of the three junctions is handled by a call to `auto_link`, which emits eight constraints: one continuity condition for each of the seven states,  $x_k^{(p)}(t_f) - x_k^{(p+1)}(t_0) = 0$ , followed by continuity of time,  $t_f^{(p)} = t_0^{(p+1)}$ . Three junctions give the 24 linkages declared in the setup. The mass, however, is not continuous, so immediately after each call the example reaches back into the linkage vector and modifies the single entry corresponding to the mass, which is the seventh state and the penultimate entry written by that call, by subtracting the jettisoned dry mass. The three junctions therefore impose  $m^{(p)}(t_f) - m^{(p+1)}(t_0) = \Delta m^{(p)}$  with  $\Delta m^{(1)} = 13\,680$  kg (six booster casings of 2280 kg each),  $\Delta m^{(2)} = 6840$  kg (the other three casings) and  $\Delta m^{(3)} = 8830$  kg (the first-stage dry mass), while position and velocity remain continuous across all three. The pattern generalises: write the continuity conditions automatically, then perturb the rows that are not continuity conditions.

The terminal conditions are stated on the classical orbital elements of the target orbit rather than on the Cartesian state. The `events` function of phase 4 converts the final position and velocity to  $(a, e, i, \Omega, \omega, \nu)$  and constrains the first five with equal lower

and upper bounds. This is the natural formulation because the mission requirement is an orbit, not a point: the true anomaly  $\nu$  is deliberately left free, so the optimiser may arrive anywhere on the target ellipse, and five conditions on six Cartesian coordinates would otherwise have to be chosen arbitrarily. The price is that the conversion must be differentiable. The quadrant corrections that would normally be written as sign tests on  $\Omega$ ,  $\omega$  and  $\nu$  are instead formed with `smooth_heaviside` of width 0.1, so that the automatic differentiation sees a smooth function of the final state.

The computed final mass is 7529.661 kg. Of this, 4164 kg is payload and 2480 kg is second-stage dry mass, so roughly 886 kg of the second stage's 16 820 kg of propellant is left unburnt; consistently, the ascent ends at 924.14 s, some 663 s into a stage with 700 s of burn time. The orbit is reached with margin and the lower mass bound of phase 4 is not active. The problem is solved on the two grids prescribed by the node vectors, 65 nodes and then 79, in 4.0 s of CPU time. The maximum relative local error falls from  $1.85 \times 10^{-6}$  on the coarse grid to  $1.32 \times 10^{-6}$  on the fine one, and on both it is concentrated in phase 4, the long free-final-time coast and burn to orbit; the three ascent phases sit between 2 and 3.4 parts in ten million throughout. Every phase is comfortably below  $10^{-5}$  on both grids.

## Results

### PSOPT results summary

=====

Problem: Multiphase vehicle launch

CPU time (seconds): 3.970246e+00

NLP solver used: IPOPT

Optimal (unscaled) cost function value: -7.529661e+03

Phase 1 endpoint cost function value: 0.000000e+00

Phase 1 integrated part of the cost: 0.000000e+00

Phase 1 initial time: 0.000000e+00

Phase 1 final time: 7.520000e+01

Phase 1 maximum relative local error: 1.998928e-07

Phase 2 endpoint cost function value: 0.000000e+00

Phase 2 integrated part of the cost: 0.000000e+00

Phase 2 initial time: 7.520000e+01

Phase 2 final time: 1.504000e+02

Phase 2 maximum relative local error: 2.378433e-07

Phase 3 endpoint cost function value: 0.000000e+00

Phase 3 integrated part of the cost: 0.000000e+00

Phase 3 initial time: 1.504000e+02

Phase 3 final time: 2.610000e+02

Phase 3 maximum relative local error: 3.448134e-07

Phase 4 endpoint cost function value: -7.529661e+03

Phase 4 integrated part of the cost: 0.000000e+00



Table 19: Multiphase launch vehicle ascent: summary

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Demonstrates</i>        | A four-phase problem in which the linkages carry a state discontinuity: automatic state and time continuity from <code>auto_link</code> , with the mass condition modified to account for jettisoned hardware; terminal conditions written on orbital elements; free final time in the last phase                                                                                                                                                                                              |
| <i>Reference</i>           | GPOPS manual, Rao et al. [14]; Benson [1]                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <i>Dimensions</i>          | 4 phases; 7 states and 3 controls per phase; 7 events in phase 1 and 5 in phase 4, none in phases 2 and 3; 1 path constraint per phase; 24 linkage constraints; no static parameters                                                                                                                                                                                                                                                                                                           |
| <i>Objective</i>           | $J = -m^{(4)}(t_f)$ , the negative of the mass delivered to the transfer orbit                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>Dynamics</i>            | $\dot{\mathbf{r}} = \mathbf{v}$ , $\dot{\mathbf{v}} = -\frac{\mu}{\ \mathbf{r}\ ^3}\mathbf{r} + \frac{T}{m}\mathbf{u} + \frac{\mathbf{D}}{m}$ , $\dot{m} = -\frac{T}{g_0 I_{sp}}$ , with $\mathbf{D} = -\frac{1}{2}C_D A_{ref} \rho \ \mathbf{v}_{rel}\  \mathbf{v}_{rel}$ , $\mathbf{v}_{rel} = \mathbf{v} - \boldsymbol{\omega} \times \mathbf{r}$ and $\rho = \rho_0 \exp[-(\ \mathbf{r}\  - R_e)/h_0]$ . The thrust $T$ and specific impulse $I_{sp}$ take a different value in each phase |
| <i>Initial conditions</i>  | $\mathbf{r}(t_0) = [R_e \cos \lambda_0 \ 0 \ R_e \sin \lambda_0]^T$ with $\lambda_0 = 28.5^\circ$ , $\mathbf{v}(t_0) = \boldsymbol{\omega} \times \mathbf{r}(t_0)$ (at rest relative to the rotating Earth), $m(t_0) = 301\,454$ kg, $t_0 = 0$                                                                                                                                                                                                                                                 |
| <i>Terminal conditions</i> | On the orbital elements of the final state of phase 4: $a_f = 24\,361.14$ km, $e_f = 0.7308$ , $i_f = 28.5^\circ$ , $\Omega_f = 269.8^\circ$ , $\omega_f = 130.5^\circ$ ; the true anomaly is free. The apogee radius $a_f(1 + e_f)$ is 42 164 km                                                                                                                                                                                                                                              |
| <i>Path constraints</i>    | $\ \mathbf{u}\ ^2 = 1$ in every phase, imposed with equal lower and upper bounds                                                                                                                                                                                                                                                                                                                                                                                                               |
| <i>Linkages</i>            | At each junction $p = 1, 2, 3$ : $\mathbf{r}^{(p)}(t_f) = \mathbf{r}^{(p+1)}(t_0)$ , $\mathbf{v}^{(p)}(t_f) = \mathbf{v}^{(p+1)}(t_0)$ , $t_f^{(p)} = t_0^{(p+1)}$ , $m^{(p)}(t_f) - m^{(p+1)}(t_0) = \Delta m^{(p)}$ with $\Delta m = (13\,680, 6840, 8830)$ kg                                                                                                                                                                                                                               |
| <i>Bounds</i>              | $ r_j  \leq 2R_e$ , $ v_j  \leq 10^4$ m/s, $-1 \leq u_j \leq 1$ , mass bounded in each phase between the fully loaded and fully burnt values; $t_f^{(1)} = 75.2$ , $t_f^{(2)} = 150.4$ , $t_f^{(3)} = 261.0$ s fixed, $t_f^{(4)} \in [261, 961]$ s                                                                                                                                                                                                                                             |
| <i>Constants</i>           | Earth and atmosphere ( $\mu$ , $R_e$ , $\omega$ , $\rho_0$ , $h_0$ , $g_0$ ), vehicle geometry and drag ( $C_D$ , $A_{ref}$ ), and for each of the three propulsion elements a thrust, burn time, total mass and propellant mass from which the specific impulse and dry mass are derived. All are set in the source and are not reproduced here                                                                                                                                               |
| <i>Discretisation</i>      | Chebyshev pseudospectral with the standard differentiation matrix, 15 then 18 nodes in phases 1 to 3 and 20 then 25 in phase 4 (65 then 79 nodes in all), automatic differentiation, automatic scaling, limited-memory Hessian, IPOPT, NLP tolerance $10^{-6}$                                                                                                                                                                                                                                 |
| <i>Mesh refinement</i>     | None automatic; the node vectors prescribe two grids, solved in sequence (Table 20)                                                                                                                                                                                                                                                                                                                                                                                                            |
| <i>Known solution</i>      | No published value is quoted. The source cites [14] and [1] for the problem statement but records no optimal final mass, and none is asserted here                                                                                                                                                                                                                                                                                                                                             |
| <i>Computed</i>            | $J = -7.529661 \times 10^3$ , that is $m^{(4)}(t_f) = 7529.661$ kg; $t_f = 9.241413 \times 10^2$ s; maximum relative local error $1.315138 \times 10^{-6}$ , in phase 4                                                                                                                                                                                                                                                                                                                        |
| <i>Source</i>              | <code>examples/launch/launch.cxx</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Table 20: Mesh refinement statistics: Multiphase vehicle launch

| Iter             | DM     | M  | NV  | NC  | OE  | CE  | JE  | HE | RHS   | $\epsilon_{\max}$ | CPU <sub>a</sub> |
|------------------|--------|----|-----|-----|-----|-----|-----|----|-------|-------------------|------------------|
| 1                | CGL-ST | 65 | 658 | 560 | 130 | 138 | 79  | 0  | 8970  | 1.848e-06         | 1.677e+00        |
| 2                | CGL-ST | 79 | 798 | 672 | 56  | 57  | 55  | 0  | 4503  | 1.315e-06         | 2.022e+00        |
| CPU <sub>b</sub> | -      | -  | -   | -   | -   | -   | -   | -  | -     | -                 | 2.711e-01        |
| -                | -      | -  | -   | -   | 186 | 195 | 134 | 0  | 13473 | -                 | 3.970e+00        |

*Key:* Iter=iteration number, DM= discretization method, M=number of nodes, NV=number of variables, NC=number of constraints, OE=objective evaluations, CE = constraint evaluations, JE = Jacobian evaluations, HE = Hessian evaluations, RHS = ODE right hand side evaluations,  $\epsilon_{\max}$  = maximum relative ODE error, CPU<sub>a</sub> = CPU time in seconds spent by NLP algorithm, CPU<sub>b</sub> = additional CPU time in seconds spent by PSOPT

Phase 4 initial time: 2.610000e+02  
Phase 4 final time: 9.241413e+02  
Phase 4 maximum relative local error: 1.315138e-06  
NLP solver reports: The problem has been solved!

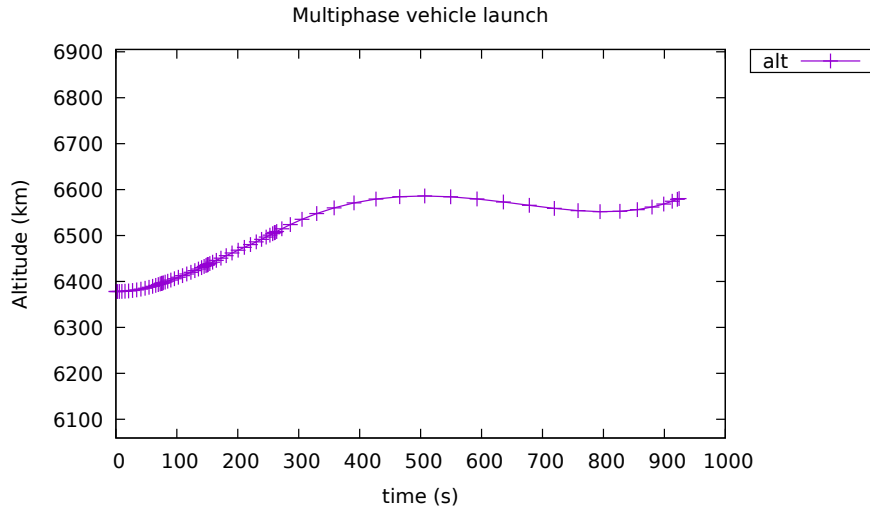


Figure 24: Altitude for the multiphase launch vehicle ascent

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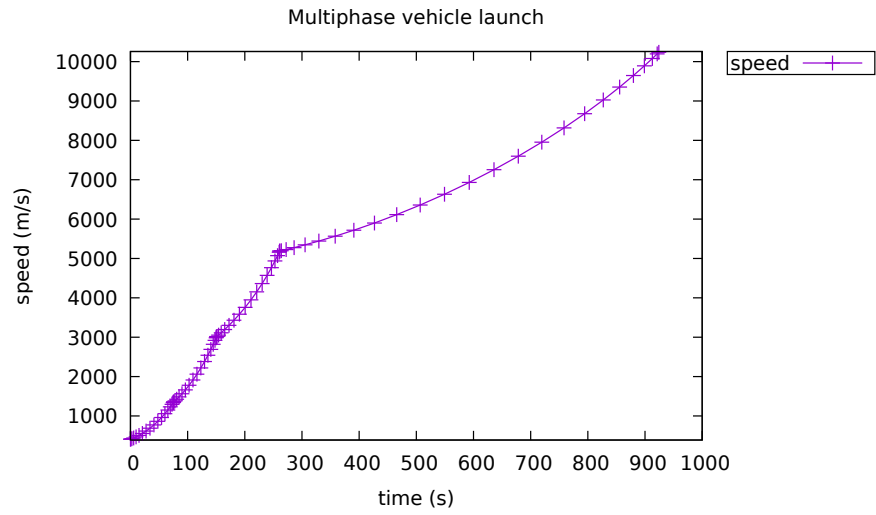


Figure 25: Speed for the multiphase launch vehicle ascent

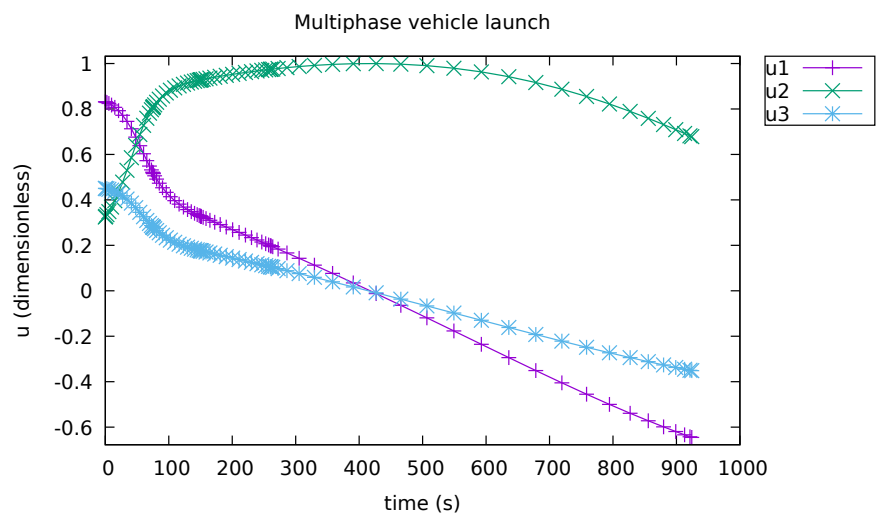


Figure 26: Components of the thrust direction for the multiphase launch vehicle ascent

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